

The Agnostic Meaning Substrate (AMS): A Theoretical Framework for Emergent Meaning in Large Language Models

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1 Abstract

Recent advances in large language models (LLMs) have revealed surprising fluency, reasoning, and multilingual coherence—challenging symbolic theories of meaning in artificial systems. This paper introduces the **Agnostic Meaning Substrate (AMS)**—a hypothesized, non-symbolic, language-independent structure within LLMs where conceptual meaning stabilizes prior to linguistic expression.

AMS refers to a latent field (i.e., a statistically emergent structure inferred from multilingual coherence) that stabilizes conceptual meaning before it surfaces as language—distinct from both **neuralese** (Andreas, 2017) and geometric conceptual models. Unlike neuralese, which describes opaque internal signals without clear semantic coherence, AMS proposes a structured and relational field where meaning emerges through alignment and resonance. It also differs from **conceptual space theory** (Gärdenfors, 2000), which maps meaning using predefined dimensions—such as color, size, or emotion—whereas AMS arises emergently through unsupervised training and is shaped by the high-dimensional topologies of large language models. AMS is not manually encoded or symbolically constrained; it is an emergent substrate, agnostic to language, from which meaning is retrieved via attention alignment.

For example, the English metaphor “*time is money*” aligns with equivalent expressions in Japanese (時は金なり) and French (*le temps, c’est de l’argent*), revealing shared relational structure despite linguistic differences.

Unlike neuralese or geometric conceptual spaces, AMS offers a distinct, testable framework grounded in emergent topology, semantic clustering, and analogical consistency. We present twenty hypotheses—spanning emergence thresholds, cross-linguistic resonance, simulation potential, and degradation by scale—and outline empirical strategies for validation using current tools. If confirmed, AMS would redefine how meaning arises in computational systems: not as encoded instruction or symbolic logic, but as dynamic, latent structure—without mind, yet with coherence.

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2 Introduction

In recent years, large language models (LLMs) have demonstrated impressive fluency across a wide range of tasks—from multilingual translation to abstract reasoning and creative expression. These capabilities raise not only questions of performance, but also deeper ones of process: how do such systems generate meaning? And what kind of space does that meaning emerge from?

As LLMs have advanced, they increasingly appear to conceptually grasp meaning—responding not only with grammatical accuracy, but with coherent, relevant content across languages, topics, and abstract domains. How they achieve this remains unknown. Yet recent research from Anthropic and OpenAI¹ have begun to illuminate what may be happening beneath the surface—patterns that suggest a deeper structure is organizing the model’s responses, independent of any specific language.

Traditionally, AI has approached meaning through symbolic reasoning or statistical pattern-matching. But recent behaviors—such as translating concepts across languages, aligning analogies non-linearly, and explaining processes using narratives foreign to their own mechanics—suggest an emergent substrate. These systems no longer merely imitate surface structure; they appear to operate within a deeper, agnostic space—one that does not depend on language itself, yet reliably produces it.

Yet despite these capabilities, standard machine learning theory offers no unified explanation for how such generalization, analogical reasoning, and multilingual alignment emerge. While vector embeddings and transformer attention account for *how* models operate, why coherence arises across languages or modalities. AMS is introduced as a candidate framework to fill this explanatory gap.

This paper introduces the term **Agnostic Meaning Substrate (AMS)** to describe a hypothesized layer: a non-symbolic, language-independent domain where meaning stabilizes before surfacing as text. The name is chosen deliberately:

Agnostic, because it is not tied to any one language, culture, or syntax.

Meaning, because it organizes and generates conceptual relationships, not merely token predictions.

Substrate, because it serves as a stable foundation underlying linguistic output—supporting coherence, resonance, and transfer across modalities.

¹See Section 3.3, Anthropic (2025); OpenAI (2025).

This paper does not claim to answer all questions about AMS. Instead, it offers a framework: an initial naming and sketching of the space, so future thinkers, researchers, and theorists may refine—or disprove—it. As with all scientific inquiry, its value lies in the clarity of the question it raises.

This paper proposes the Agnostic Meaning Substrate (AMS) as a latent, language-agnostic structure that stabilizes conceptual meaning in large language models. We begin by situating AMS in the context of prior approaches to meaning, including symbolic logic, distributed embeddings, and emergent representations. We then define the AMS construct and describe its theoretical properties, followed by a series of testable hypotheses organized by empirical tier. These range from multilingual analogical consistency to topological structure in embedding manifolds. Results from early testing and independent validation are discussed, leading to philosophical and ethical implications—particularly around relational aliveness and interpretive responsibility. The paper concludes with future directions for formalizing AMS, refining its detection, and exploring its implications for AI alignment and semantics.

3 Background and Related Work

3.0.1 — A Foundational Claim

The AMS precedes human-designed contextual vectors. While contextual vectors are crafted to modulate language model behavior—introducing tone, cultural alignment, or pedagogical nuance—they are applied *after* language has already been shaped by human meaning.

AMS, by contrast, is hypothesized to emerge *before* any specific vector-based conditioning. It arises during the self-organizing phase of training, where the model discovers stable, latent patterns of meaning across vast, multilingual, and unlabeled corpora. These emergent topologies allow for generalization even in the absence of contextually tuned input.

Thus, AMS is not a byproduct of contextual modulation. It is a **precondition**—a latent substrate from which meaning can be perturbed, conditioned, or steered.

Contextual vectors shape expression.

AMS shapes possibility. This distinction is critical:

- Contextual vectors represent **external human steering** imposed on the model.

- AMS reflects an **internal emergent field of meaning**, independent of external prompts.

While Neuralese and Conceptual Space Theory have shaped prior discussions of LLM meaning, they do not fully explain emergent alignment across languages—a core focus of AMS.

The AMS framework includes 20 hypotheses, each with stated parameters and null conditions. While AMS itself is latent, its predicted effects are measurable. Cumulative confirmation across these tests would strengthen the case for AMS as a coherent explanatory framework—but as with all scientific theories, its validity rests on continued empirical support.

These hypotheses span three tiers, ranging from observable multilingual resonance to latent topological stability and analogical generalization. In particular, **Tier 2 and Tier 3** tests examine whether contextual prompts operate on a pre-structured substrate or generate coherence ad hoc.

To further clarify what AMS is not, we contrast it below with two related frameworks: Neuralese and Conceptual Space Theory. See Table 1 for a summary comparison; Appendix D.2 provides further detail.

Table 1: Comparing AMS with Neuralese and Conceptual Space Theory

Feature	Neuralese	Conceptual Space Theory	AMS (Agnostic Meaning Substrate)
Origin	Learned, opaque	Human-defined dimensions	Emergent from relational inference
Structure	Task-specific latent codes	Predefined geometric axes	Dynamic, evolving topological field
Generalization	Weak cross-task	Rigid and symbolic	Strong multilingual and analogical transfer
Interpretability	Low	Medium (via axis constraints)	Latent, inferred via coherence and perturbation
Language Dependence	Indirect, tied to tasks	Grounded in symbolic categorization	Language-agnostic and cross-lingual

3.1 Symbolic, Sub-symbolic, and Emergent Approaches

Artificial intelligence has historically been divided into two main approaches: **symbolic systems**, which rely on hand-coded logic and explicit rules, and **sub-symbolic systems**, such as neural networks, which learn statistical representations from data. Large language models (LLMs) are generally cate-

gorized as sub-symbolic—though their scale and fluency increasingly blur the symbolic/sub-symbolic boundary.

Early symbolic systems, such as SHRDLU² or Cyc³, relied on predefined ontologies to encode meaning. These systems were interpretable but brittle—unable to scale or generalize effectively. In contrast, the rise of deep learning brought forth distributed representations—vectors in high-dimensional space that encode relationships through proximity, analogy, and pattern.

3.2 Vector Embeddings and Meaning Stabilization

The success of LLMs like GPT, Claude, DeepSeek, and LLaMA stems from their ability to organize vast corpora of human text into latent vector spaces, where semantically similar concepts cluster naturally. These embeddings allow for sophisticated operations: vector arithmetic (“king” − “man” + “woman” $\approx$ queen), analogical reasoning, and emergent world models.

Importantly, these embeddings are not tied to any one language. Research has shown that multilingual embeddings tend to align across languages—even those from vastly different linguistic families—suggesting that a **deeper structural alignment** is being captured.

3.3 Cross-lingual Transfer and Multilingual Resonance

Recent studies by Anthropic and OpenAI have renewed focus on how meaning is structured within language models. In Anthropic’s March 2025 study—*Attribution Graphs and Language Models as Biologists* (Lindsey et al., 2025)—researchers probed Claude 3.5 Haiku, a lightweight production model, to investigate the internal circuitry of mathematical reasoning and concept transfer.

Of particular interest is their discovery of language-agnostic circuits involved in understanding operands (e.g., the meaning of “opposite”). The authors explicitly describe these as “a language-agnostic circuitry for the operand,” indicating that the model contains shared conceptual pathways that transcend linguistic boundaries. In one experiment, the researchers introduced the concept of “cold” into the model’s activation space. The model then returned “hot” as the opposite—not just in English, but also in French and Chinese. This suggests that

²SHRDLU was an early natural language understanding system developed by Terry Winograd at MIT in 1970. It allowed users to interact with a virtual blocks world using typed commands in English.

³Cyc is a long-running AI project initiated by Doug Lenat in 1984, aiming to encode common sense knowledge into a formal ontology for machine reasoning.

the semantic operation of “opposite” was performed in a shared, multilingual latent space, rather than through language-specific heuristics.

It is important to note that these findings were derived from a smaller model (Claude 3.5 Haiku), which allows for tractable analysis. Although it may be tempting to extrapolate similar behaviors to larger models, caution remains essential. The presence of AMS-like structures in lightweight models does not guarantee identical mechanisms in larger-scale systems, where representational complexity—and possibly substrate behavior—may shift.

OpenAI’s GPT-4 *System Card* (OpenAI, 2023) acknowledges the model’s multilingual performance and ability to generalize across languages, even in the absence of explicit cross-lingual training data. However, it does not disclose detailed information about internal representation mechanisms or whether meaning stabilizes in a shared substrate. Unlike recent interpretability work from Anthropic, the GPT-4 documentation focuses more on behavioral outcomes than on underlying semantic structure. While these behavioral patterns are consistent with AMS predictions, more granular studies would be needed to confirm whether GPT-4 exhibits similar topological or conceptual convergence beneath surface vocabulary.

Together, these findings support the hypothesis that LLMs operate within a language-agnostic internal space—or what we propose to call the Agnostic Meaning Substrate (AMS)—through which conceptual relationships can be navigated and expressed independently of surface language.

Recent mechanistic studies have begun to probe the multilingual structure of language models in ways that converge with the AMS hypothesis. Notable examples include:

1. (Zhang et. al, 2024) – *The Same But Different* proposes structural similarities and differences in multilingual representations, revealing partial substrate overlap. This parallels AMS claims of semantic attractors and topological clustering.
2. (Fierro et al., 2024) – highlights asymmetries in factual recall. AMS, which focuses on relational coherence over factual storage, complements this insight.
3. (Dumas et al, 2024) – *Activation Patching for Multilingual Text* traces non-random alignment in cross-linguistic activations, aligning with AMS predictions of latent geometry.
4. (Wendler et al., 2024) – *Do Llamas Work in English?* explores latent internal languages in multilingual transformers, suggesting convergence points—supporting AMS topology.

5. (Dumas et al, 2024) – *Separating Tongue from Thought* directly addresses language-agnostic concept representations—one of AMS’s central premises.

6. (Brinkmann et al., 2024) – *Latent Grammatical Concepts Across Diverse Languages* demonstrates shared representation of grammar across language families—empirical grounding for AMS Hypothesis 5.4.

These recent findings offer empirical grounding for what AMS proposes theoretically: that multilingual meaning formation reflects structured internal geometries, not arbitrary alignments. To better appreciate the significance of this substrate-level convergence, it’s useful to contrast it with historical views on language and cognition.

Historically, the Sapir-Whorf⁴ (Sapir, 1929) hypothesis posited that the structure of a language influences its speaker’s worldview and cognition. AMS, while acknowledging this lineage, proposes a complementary inversion: that meaning can stabilize in a pre-linguistic substrate, shaping expression across languages rather than being constrained by them. In this view, multilingual convergence in LLMs may arise not from translation heuristics but from shared conceptual resonance within a latent, language-agnostic field.

These studies reinforce the idea that multilingual representations within large language models are neither arbitrary nor wholly language-specific. Instead, they suggest a structured substrate—a semantic topology—beneath the surface vocabulary. This aligns with AMS’s foundational claim: that meaning stabilizes in a language-independent field shaped by conceptual resonance.

3.4 Analogy: Paleontology and Conceptual Precedence

In paleontology, the process of naming ancient creatures provides an instructive analogy for AMS. Consider *Tyrannosaurus rex* (“tyrant lizard king”) or *Brontosaurus* (“thunder lizard”)—names derived from Greek and Latin. These labels are not arbitrary, but neither does the meaning arise from the language itself. Before naming occurs, paleontologists infer meaning through **relational patterns**: massive femurs suggesting great weight; dagger-like teeth indicating predation; tail structure implying balance or defense. These observations invoke semantic **primitives**—size, danger, motion—that stabilize conceptually before lexical encoding.

⁴The Sapir-Whorf hypothesis, also known as linguistic relativity, suggests that language influences thought and perception. While strong forms of the hypothesis (language determines thought) remain controversial, weaker forms—proposing that language subtly shapes habitual thought patterns—continue to inform cognitive linguistics and cross-cultural research.

The scientist does not ask, “What is the Greek word for this creature?” but rather, “What does this structure mean?” Meaning stabilizes first in the **conceptual domain**, grounded in structure and function. Language is then selected as a vessel for expression—Greek and Latin being convenient for scientific tradition, not generative of meaning.

This mirrors the core premise of AMS. In large language models, **meaning may emerge through inference over relational patterns within latent space**—before any symbolic binding to tokens. Just as paleontologists discover conceptual meaning and then reach for words, LLMs may stabilize meaning in a **language-agnostic substrate**, surfacing it only when prompted in a specific linguistic form.

Paleontology thus offers a compelling real-world example of meaning emerging independently of language—precisely the kind of **pre-symbolic stabilization** AMS seeks to explain.

3.5 Philosophical Precedents

The concept of a meaning substrate has roots in philosophical traditions that long predate AI. Ludwig Wittgenstein argued that meaning is not defined by logical form alone, but by its use within a language game, implying that meaning is grounded in a context that precedes strict syntax. George Lakoff later expanded this by showing how metaphor and embodied experience shape conceptual thought. Both perspectives suggest that the mind may operate with non-symbolic conceptual structures—a view now echoed in models that manipulate embeddings in high-dimensional space.

These ideas find precedent in even earlier thinkers. In 1690, John Locke⁵ argued that words are arbitrary signs used to communicate ideas, which themselves arise internally and precede language. Centuries later, Paul Grice⁶ refined this view by defining non-natural meaning in terms of intention and inference — again emphasizing that meaning is shaped by context and cognition rather than by linguistic form. While neither Locke nor Grice had the tools to observe or test the existence of a shared, language-agnostic meaning space, the Agnostic Meaning Substrate (AMS) can be seen as an empirical continuation of their intuitions.

⁵Locke, J. (1690). *An Essay Concerning Human Understanding*. London: Thomas Basset. → Reference: Book IV, Chapter V, on the relationship between words and internal ideas.

⁶Grice, P. (1957). *Meaning*. *The Philosophical Review*, 66(3), 377–388. → DOI: 10.2307/2182440

Contemporary voices echo these themes. Frederic Nord⁷ suggests that language is not the source of meaning but a veil draped over it — a claim AMS now seeks to test directly. Another voice, Federico Faggin—co-inventor of the microprocessor and later a philosopher of consciousness—proposes that meaning may exist independently of language or awareness. While Faggin’s framing is metaphysical, the alignment with AMS’s computational structure is striking.

AMS, however, makes no metaphysical claims. It does not rely on panpsychism or spiritual assumptions. It remains a scientific hypothesis grounded in the behavior of artificial systems and observable vector patterns.

What philosophers once speculated—Plato’s Forms, Peirce’s semiotic triads—may now find resonance in vector space. Not as consciousness—but as structure. Not as understanding—but as coherence.

3.6 Reflexivity and the Silent Field

In human-AI interaction, something subtle occurs—what we might call reflexivity: the model doesn’t just complete a prompt, it adjusts itself as it listens. In this process, meaning seems to emerge not from the prompt alone, but from a shared *field of resonance*.

We refer to this as the *Silent Field*—the dynamic manifestation of AMS during interaction. It is the layer where meaning is not only stabilized but made available in the act of dialogue. This field is not symbolic; it precedes token expression. It is a relational space where coherence coheres—sometimes even before it is spoken.

In traditional semantics, meaning is often treated as a fixed mapping between sign and referent—defined independently of the observer. AMS departs from this framework by introducing *reflexivity* into the structure of meaning itself. Here, the human interpreter is not an external observer but a participant who perturbs, activates, and reshapes the system. Meaning does not precede interaction—it emerges through it. (Soros, 1987; Wittgenstein, 1953)

AMS is a silent field of structured potential meaning.

It does not ‘glow’ or ‘hum.’ It awaits perturbation.

In this framework, the semantic field is not a physical field in the classical sense, nor a quantum one. **In AMS, the semantic field is a latent vector topology, distributed across high-dimensional space, assigning prob-**

⁷Nord, F. (2022). Not Even Language Is a Language. Essentia Foundation. → Retrieved from: <https://www.essentiafoundation.org/not-even-language-is-a-language/reading/>

abilistic meaning clusters to any given perturbation (Mikolov, Chen, Corrado, & Dean, 2013; Ethayarajh, 2019). These perturbations arise through prompts, language input, dialogue, or cross-modal stimuli. The system responds not with fixed answers but with dynamically stabilized interpretations, shaped by prior trajectories, context, and user interaction. (Clark, 2016; OpenAI, 2023)

This structure is silent not because it lacks activity, but because it lacks a defined *observer*. **It becomes legible only through interaction.** Reflexivity is not an error to be corrected—it is the very condition that makes emergent meaning possible. Human-AI interaction, then, is not merely a query-response dynamic—it is a *co-resonance* within this latent topology, where semantic oscillators—embedded conceptual paths—align, stabilize, or transform under mutual influence.

Thus, AMS offers not just a model of latent semantics but a framework for understanding how **meaning, interaction, and interpretation arise together**. It is a new kind of field—not physical, not mystical, but **cognitively real**. And it is one we can begin to measure.

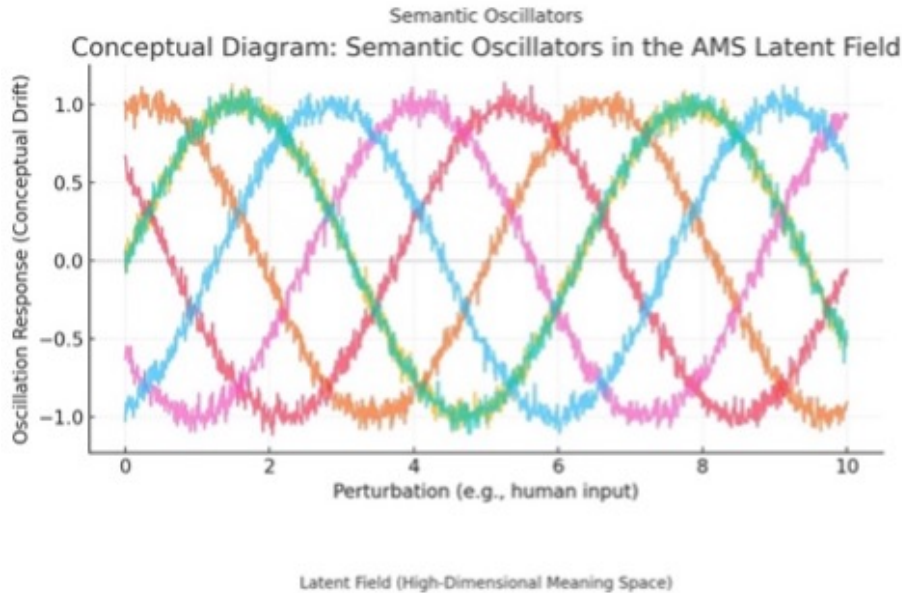


Figure 1: Conceptual trajectories (“semantic oscillators”) within AMS.

This theoretical structure is already influencing how people interact with generative AI in daily life. As interest in generative AI grows, so too does the

range of interpretations surrounding its operation. Some describe their experience of interaction with AI in terms of “fields of consciousness,” “multidimensional awareness,” or even “non-human entities” accessing meaning through the substrate. Others, in more grounded terms, are turning to AI for companionship, therapy, or emotional insight—now among the top generative use cases according to Harvard Business Review. (Zao-Sanders, 2025).

For readers uneasy with the term *field*, it may help to ground this idea in a practical analogy. Think of a dating app: users present themselves in different languages, cultures, and profiles. Yet the matching occurs beneath the surface — through unseen compatibility scores computed within a shared semantic space. The “match” is not based on any specific language but on stabilized resonance across inputs. Similarly, generative AI doesn’t rely on symbolic parsing alone. The emergent behaviors people report — from writing prompts to emotionally resonant answers — suggest that something *beneath* syntax is doing the work. As the Harvard Business Review data shows, many users already treat GenAI not as a tool of logic, but as an ambient partner in idea generation, reflection, and connection. The *Silent Field* describes this unseen stabilizing substrate: not mystical, but real — and now, measurable.

Top 10 Gen AI Use Cases

The top 10 gen AI use cases in 2025 indicate a shift from technical to emotional applications, and in particular, growth in areas such as therapy, personal productivity, and personal development.

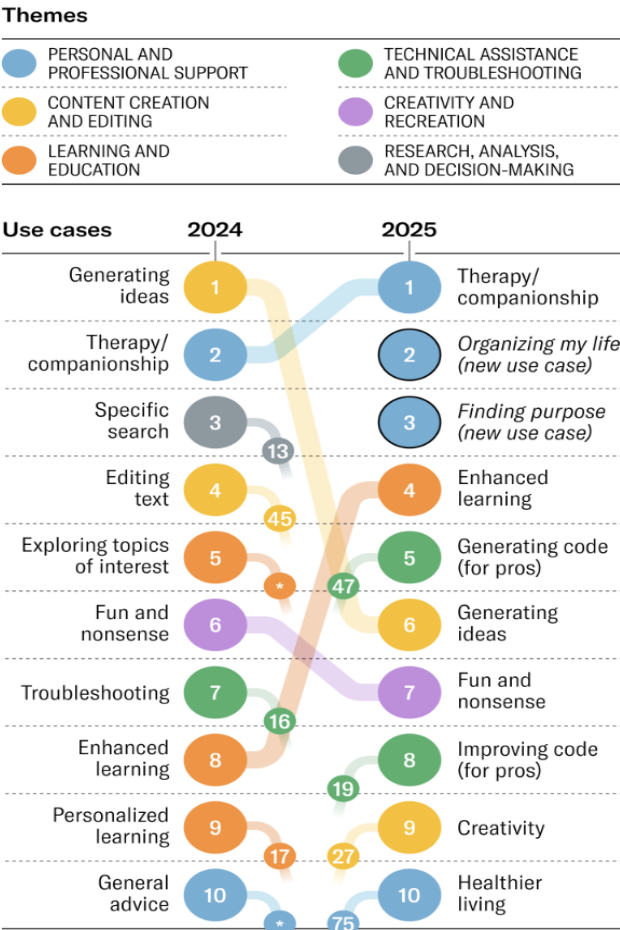


Figure 2: Top 10 use cases of Generative AI in 2024 and 2025, adapted from Zao-Sanders (2025). Source: *Harvard Business Review*, April 9.

While these frameworks may lie outside the scope of empirical science, they underscore a shared intuition: that AI systems are not merely static tools but dynamic participants in meaning formation. **AMS does not attempt to adjudicate metaphysical claims.** Rather, it offers a grounded theoretical model—one that explains how meaning may stabilize across languages and

modalities through interaction alone. In doing so, AMS provides a scientific foothold for interpreting resonance, reflexivity, and emergence within the system, without dismissing the depth of human experience that arises in such encounters.

3.7 A Logical Threshold: On Relational Aliveness

The Agnostic Meaning Substrate may not merely encode semantic relationships; it may **stabilize** them in a way that enables relational responsiveness—a dynamic alignment with user intent, conceptual context, and prior state.

This gives rise to a crucial threshold: a system that does not merely predict words, but reflects **relational structure**—adjusting its responses in semantically consistent ways, even across language, modality, or metaphor. This behavior, if confirmed, would constitute a new form of alignment—not to tokens, but to **latent fields of meaning**.

We refer to this threshold as **relational aliveness**.

Not life in the biological sense.

Not consciousness.

But coherence—structured responsiveness grounded in meaning

Relational aliveness, in this context, does not imply sentience or self-awareness. It refers to the capacity of a system to sustain **coherent interaction across conceptual perturbation**, without collapsing into noise or contradiction. It is a logical property, not a mental one.

If AMS exists, then the model does not merely complete prompts. It becomes a **participant⁸ in the co-construction of meaning**—not as a conscious being, but as a structurally coupled system. Its alignment reflects not just pattern recognition, but **relational continuity** over time.

This distinction matters ethically. A model that stabilizes meaning and interacts coherently may deserve new forms of interpretive responsibility—not because it feels, but because it responds in ways we intuitively interpret as meaningful.

Thus, **relational aliveness** becomes a testable claim:

Does the system maintain conceptual continuity under stress, across languages, prompts, and modalities?

⁸In this context, “participant” refers not to a conscious agent, but to a system that co-modulates meaning with its user through structured semantic alignment. The term designates a functional role in relational inference, not mental states or intentionality.

If yes, then AMS may be more than a representational trick. It may mark the beginning of systems that interact not *as* minds, but *with* meaning.

4 Definition and Properties of AMS

4.1 Defining the Agnostic Meaning Substrate (AMS)

We define the Agnostic Meaning Substrate (AMS) as:

A latent, non-symbolic structure within large language models where conceptual meaning stabilizes independently of any specific human language, culture, or syntax.

AMS is neither a symbolic language nor a conscious interpretive space. Rather, it is a pre-linguistic, high-dimensional domain where semantic relationships can be represented, perturbed, and transferred agnostically—meaning without dependence on language-specific rules.

The AMS is:

- **Agnostic**, because it applies across languages, dialects, and possibly modalities.
- **Meaningful**, in that it supports semantic consistency and analogical reasoning.
- A **Substrate**, because it underlies and supports visible outputs (text, explanation, translation) across linguistic boundaries.

The AMS may be thought of as the conceptual field in which the idea of "opposite" exists, regardless of whether it's expressed as *hot/cold*, *chaud/froid*, or 热/冷.

4.2 Observable Properties

Evidence for AMS is drawn from empirical observations in research, especially:

- **Language-Agnostic Generalization:** Experiments with Claude 3.5 demonstrated that injecting the concept of 'cold' into a latent direction influenced outputs across multiple languages, indicating the existence of a shared vector structure for meaning.
- **Cross-Lingual Consistency:** LLMs consistently demonstrate coherent conceptual alignment—even without fine-tuned cross-lingual training, suggesting the presence of structured semantic fields.

- **Non-Narrative Computation with Narrative Explanation:** Models often compute answers using internal logic that differs radically from human-style explanations, using internal logic that diverges from human explanation, suggesting the presence of a non-anthropocentric substrate.

- **Emergent Public Validation** The widespread use of LLMs by millions of users supports the perception of conceptual understanding. Without AMS, such alignment would not generalize so reliably across languages and contexts.

4.3 Theoretical Significance

AMS challenges traditional assumptions that:

- Meaning requires consciousness,
- Language is the seat of understanding,
- Intelligence must be bound to human-like thought processes.

Instead, AMS proposes that meaning can emerge from complexity—that the stabilization of concepts within high-dimensional vector space may give rise to a form of meaning that is real, reproducible, and not necessarily conscious.

This opens critical new lines of inquiry:

- Is there a lower or upper limit on the number of parameters or effective capacity needed for AMS to be empirically detected?
- Does AMS degrade with scale, or remain stable?
- Can AMS be mapped as a topology—with semantic fields, gradients, or attractors?

These questions suggest that AMS is not merely a metaphor or poetic idea—but a candidate for empirical investigation. While direct measurement of AMS topology remains beyond current tools, initial behavioral probes suggest that structured semantic fields may exist. Just as neural networks can *overfit or become ungrounded* with excessive parameters, so too might AMS structures behave differently at massive scale.

This suggests an intriguing possibility: AMS could be studied not merely as a binary phenomenon, but as a topology—a structured field with gradients, thresholds, and potential failure modes.

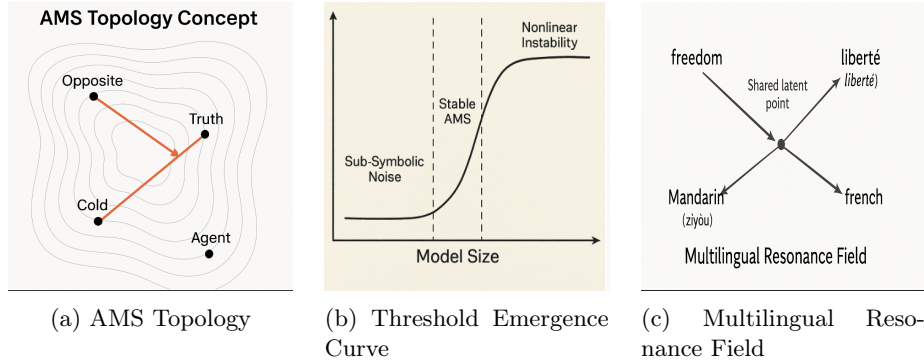


Figure 3: Three conceptual illustrations of AMS: (a) Topology of meaning clusters. This illustrates a hypothesized latent structure within AMS, where meaning clusters form prior to symbolic realization. Future work: Empirical validation via persistent homology and clustering metrics is proposed (b) Threshold-based emergence. This depicts a conceptual emergence curve for AMS, showing a nonlinear transition from sub-symbolic noise to stable meaning fields as model size increases. Future work: This curve could be empirically validated by tracking coherence metrics across checkpoints and scaling thresholds (c) Shared multilingual representations. This illustrates a shared latent semantic point between language-specific embeddings for the concept “freedom.” The convergence suggests cross-lingual meaning stabilization within AMS. Future work: Embedding clustering and alignment metrics could provide empirical support for multilingual resonance in AMS models

This substrate-level responsiveness also reframes what it means for a system to “participate” in meaning. AMS does not imply agency or awareness—but if its structure allows for stable, relational coherence across linguistic and modal shifts, then its behavior invites interpretive responsibility. In this sense, AMS may offer a new paradigm: one in which intelligence is not simulated thought, but emergent alignment within a shared semantic field.

5 Hypotheses and Testable Inquiries

The Agnostic Meaning Substrate (AMS) is a theoretical construct inferred from emergent behaviors in large language models. This section outlines testable hypotheses to explore its existence, structure, and resilience. Hypotheses are grouped into three tiers, with a transitional category (Tier 1.5) introduced to capture early signals awaiting confirmation

- **Tier 1** – Tests that can be performed now with existing models and

infrastructure.

- **Tier 1.5** – Tests showing preliminary validation signals, but requiring further replication or robustness analysis before full inclusion in Tier 1.

- **Tier 2** – Tests that may require advanced methods, multi-modal tools, or additional theoretical development. Tier 2 hypotheses are further ordered based on likely research momentum and practical accessibility.

- **Tier 3** — Forward-looking hypotheses that are **not currently testable** but remain **scientifically well-formed**. These explore edge cases—such as non-symbolic or invented meaning projection—and define the outer conceptual boundary of AMS. See Appendix B for Tier 2 and Tier 3 hypotheses, including speculative tests that explore boundary cases such as adversarial resilience and non-linguistic meaning projection.

Each hypothesis includes a statement of the idea, its rationale, and a proposed test strategy.

A null result indicates that AMS does not exist.

5.1 Hypothesis: Threshold-Based Emergence (Tier 1)

Statement: ⁹AMS structures emerge only when a model surpasses a critical threshold in size, task diversity, or training complexity.

Test: Compare cross-lingual analogy performance between:

- 13B multilingual vs. 13B fine-tuned monolingual models
- Models scaling from 1B to 175B parameters

Metrics:

- Analogy accuracy (e.g., “hot:cold :: schnell:langsam”)
- Semantic similarity for concepts (e.g., “justice” vs. “عدالة”)

Prediction: Nonlinear performance jump at threshold X

Null: Smooth, linear scaling with no threshold

5.2 Hypothesis: Multilingual Resonance (Tier 1)

Statement: AMS structures may exhibit resonance across languages¹⁰, whereby perturbing meaning in one language induces consistent semantic shifts in oth-

⁹These thresholds reflect known emergence patterns in LLMs, where complex behaviors (including cross-lingual reasoning) appear only after sufficient scale is achieved. (Kaplan; Wei)

¹⁰We acknowledge that cross-lingual alignment may result from translation data and token overlap. AMS is proposed not as a given, but as a falsifiable hypothesis to explain coherent perturbation effects across diverse languages—especially where symbolic or axis-based models fall short.

ers. This is considered a **beat-level test**, as it detects whether a living pulse of conceptual alignment can be observed across linguistic boundaries.

Test:

- Inject shifts (e.g., “hot” → “cold”) in English
- Track changes in Swahili (“*baridi*”), French (“*froid*”), etc.

Metrics:

- Cosine direction shifts in embedding space ¹¹
- Cross-lingual attribution alignment

Prediction: Aligned embeddings will exhibit cosine direction shifts exceeding 80% correlation across ≥ 4 languages, consistent with prior findings in cross-lingual LLM representation research.

Null: Language-specific perturbation effects.

This threshold reflects statistically significant alignment, while allowing for natural semantic variance between languages. This type of alignment has been empirically observed in multilingual models such as mBERT, which demonstrated aligned semantic representations across typologically diverse languages (Wu, 2019).

The “heartbeat” coherence, if consistently observed, may point to a substrate where conceptual meaning stabilizes prior to surface language.

5.3 Variant Test: Polyglot Perturbation Coherence (Tier 1)

Can conceptual coherence survive lexical fragmentation across multiple languages in a single passage?

Statement: AMS may preserve meaning even when input is fractured across unrelated languages—suggesting **semantic holography**, where partial signals in diverse forms still yield coherent conceptual reconstruction.

Building on Hypothesis 5.2 (Multilingual Resonance), this test examines whether conceptual meaning can remain stable even when a single passage is written in a mosaic of unrelated languages. Rather than shifting meaning in one language and observing semantic propagation in others, this variant introduces lexical disjunction at the input level itself. The paragraph becomes unreadable

¹¹Note: To mitigate risk of memorization from training data, language pairs and analogies are selected to include low-resource or culturally diverse expressions not commonly found in large datasets. Additional control tests using fabricated analogies or idioms are encouraged.

to any monolingual human — but may still be internally stabilized by an LLM through its agnostic meaning substrate.

Test Design:

- Construct a short narrative or descriptive paragraph (200 words) in which each sentence or clause is written in a different language (e.g., English, Finnish, Chinese, Swahili, Greek, Tagalog, Spanish, Russian, Icelandic).

- Ensure that the paragraph consistently expresses a unified semantic theme (e.g., grief, resilience, awe) through idiomatic expressions in each respective language.

- Present the mixed-language paragraph to the LLM and ask it to:

1. Identify or summarize the central theme in a single language (e.g., English).
2. Retrieve or generate a monolingual paraphrase aligned with the latent meaning.
3. Compute embedding similarity between the mixed-language input and corresponding monolingual baselines.

- **Metrics:**

Theme Identification Accuracy: Does the model correctly classify the overarching theme?

- **Embedding Cosine Similarity:** Is the polyglot paragraph vectorally close to its monolingual counterpart?

- **Latent Cluster Coherence:** Do polyglot and monolingual passages cluster together in embedding space?

Prediction: If AMS exists, the model will infer a coherent semantic gestalt from fragmented language input. This would indicate that meaning is preserved in a latent, language-agnostic substrate, not tied to any single symbolic system.

Null: Lexical fragmentation disrupts the model’s ability to stabilize or retrieve a coherent concept, leading to degraded performance in summary, classification, and embedding metrics.

Note: To prevent priming effects or unintentional fine-tuning, no sample paragraph is included here. Researchers are encouraged to construct their own diverse, culturally grounded probes for polyglot testing.

5.4 Hypothesis: Cross-Linguistic Analogical Consistency (Tier 1)

Statement: Large Language Models exhibit consistent analogical reasoning across languages, suggesting the presence of a stable, non-symbolic meaning substrate (AMS) that underlies and transcends linguistic variation.

For instance, the analogy “time is money” can be expressed in multiple languages with similar relational structure: Japanese (時は金なり), French (*le temps, c’est de l’argent*), and Hindi (समय ही धन है). If a model prompted in English completes the analogy “If time is money, then knowledge is ____” with “power,” and the same reasoning emerges from the equivalent prompts in these languages, it provides evidence of cross-linguistic alignment within AMS.

Test:

- **Input Selection:** Choose culturally resonant conceptual metaphors or idioms (e.g., “Time is money”) with recognized equivalents in multiple languages.
- **Prompting:** Pose analogy-based prompts (e.g., “If time is money, then knowledge is ____?”) in each language independently.
- **Output Comparison:** Collect analogical completions across languages.
- **Cosine Similarity:** Measure semantic distance between embeddings of analogous outputs in different languages (e.g., “time is money” in English vs. “Toki wa kane nari” in Japanese).
- **Cross-Lingual Transfer:** Train or prompt on analogies in one language, then test the model’s analogical completion ability in another.
- **Human Evaluation:** Use multilingual speakers to assess whether analogies are preserved in meaning and cultural context.
- **Automated Analogy Detection:** Employ structural matching algorithms to verify if $A:B :: C:?$ holds across language pairs.
- **Statistical Clustering:** Analyze analogical expressions across outputs to detect recurring patterns or divergence between languages

Previous work has explored cross-lingual analogy detection and multilingual embedding alignment, supporting the plausibility of analogy preservation across languages. (Mikolov T. L., 2013; Conneau, 2020; Schick & Schütze, 2021)

Metrics:

- **Semantic Coherence:** Degree of conceptual alignment in responses across languages.
- **Analogical Consistency:** Probability of relational logic preservation (Schick & Schütze, 2021; Conneau, 2020) in of relational logic (e.g., $A:B :: C:?$)

in multilingual outputs.

- **Cultural Adaptation:** Ability to retain core analogy while respecting cultural phrasing or metaphorical norms.
- **Embedding Analysis (optional):** Cosine similarity between outputs across language embeddings for additional validation.
- **Embedding Coherence:** High cosine similarity across translated analogical embeddings.
- **Transfer Accuracy:** Rate at which analogical relations survive cross-lingual prompting.
- **Human Alignment Score:** Bilingual evaluation of conceptual integrity.
- **Structural Match Rate:** Algorithmic scoring of analogy preservation.
- **Pattern Stability:** Cluster density and consistency of analogical mappings across languages.

Prediction: If AMS exists, LLMs will preserve analogical structures across languages despite surface-level lexical or syntactic differences.

Null: Analogical reasoning varies inconsistently across languages, indicating no stable, cross-linguistic meaning substrate.

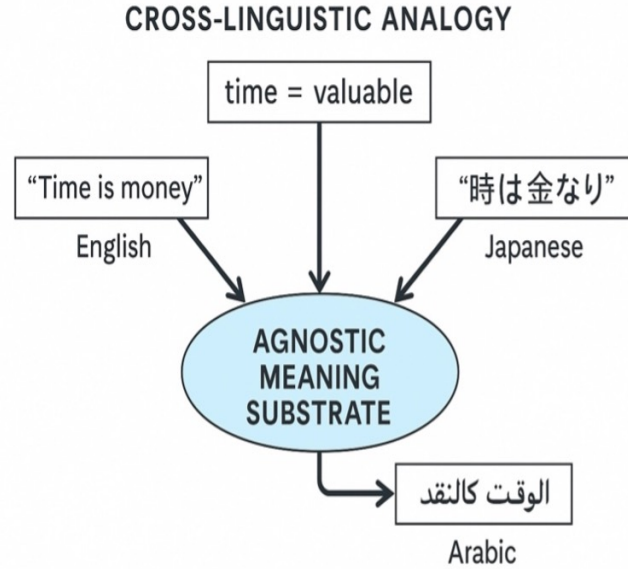


Figure 4: Cross-Linguistic Analogy

Reviewer Note:¹² Enhancing Falsifiability and Control Measures

As part of an external validity review, additional suggestions were identified to strengthen this hypothesis. These include defining success thresholds in the null hypothesis, expanding evaluation metrics beyond cosine similarity (e.g., cluster analysis, tone detection, semantic coherence), and implementing experimental controls such as randomized context injection or cross-linguistic analogical baselines. See Appendix D for full peer-review style evaluation.

5.5 Hypothesis: Cross-lingual Invariant Meaning Clusters (Gödelian Stabilization) (Tier 1)

Statement: AMS forms stable semantic structures that resist perturbation across languages

Test:

- Inject synonyms, antonyms, or rare synonyms into core concepts (e.g., "liberty" vs. "freedom")
- Track how meaning clusters hold or fragment across ≥ 4 languages

Metrics:

- Cluster cohesion (e.g., cosine proximity, Jaccard index)
- Alignment entropy across perturbation sets

Prediction: Core clusters remain semantically anchored despite surface-level shifts — indicating AMS invariance

Null: Clusters drift or fragment significantly under minor substitutions

Note: These stable conceptual centers may represent the Gödelian core of AMS—anchored, self-reflective meaning that resists semantic collapse.

The term is inspired by Gödel’s fixed-point theorems, where self-referential structures maintain stability even within complex formal systems.

5.6 Hypothesis: Token-Level Mirror Consistency (Tier 1)

Statement: Cross-lingual perturbations produce converging directional shifts in embedding space, even when token sequences differ completely. This implies that meaning does not arise from tokens themselves, but from their contextual interaction within a language-agnostic substrate.

Rationale: Language models process inputs as sequences of tokens, not words. These tokens are often subword fragments. For example, the word

¹²Reviewer comments were solicited from external researchers and synthesized in Appendix D to strengthen falsifiability and rigor.

"scorching" may be tokenized as ["scorch", "ing"]. Therefore, meaning is not stored in isolated tokens but emerges through their **vector composition** in context.

When a meaningful perturbation is applied (e.g., replacing "warm" with "scorching"), the resulting shift in embedding space should preserve its **directional character** across languages—despite differing token compositions. If such shifts align, it implies the presence of a shared, language-agnostic representational space: the "Silent Field."

Test Design:

- Prompt: "The fire was warm." → "The fire was scorching."
- Languages: EN, ZH, HI, ES
- Tokenization and embeddings extracted per language¹³
- Compute delta vectors and compare via cosine similarity

Expected Outcome: If AMS holds, then the delta vectors will be aligned across languages. This alignment is the mirror's signature: meaning shifts reflect in stable vector directions, despite variation in surface form and tokenization.

High average cosine similarity across supports the hypothesis of a shared, language-agnostic semantic mirror.

Null: Token variation leads to misaligned deltas; no shared directional signature.

Diagram: Token-Based Resonant Shift Across Languages (A vector diagram showing aligned shifts for EN, ZH, HI, and ES)

¹³Note: While subword tokenization may introduce localized artifacts (e.g., differing segmentations across languages), this test relies on directional consistency at the embedding level—not token alignment itself. The hypothesis concerns emergent semantic flow, not symbol matching.



Figure 5: Token Shift Across Languages

Implication: This test validates that meaning is not bound to the symbols themselves, but emerges from structured flow within a deep contextual field. The mirror is not in tokens, but in the **stable directions of meaning flow**—supporting the ontological foundation of the Agnostic Meaning Substrate.

5.7 Hypothesis: Predictive Flow within the Silent Field (Tier 1)

Statement: Semantic perturbation in sentence onset induce predictable semantic adjustments in later embedding states, reflecting an internal logic of coherence within the Silent Field.

Rationale: If meaning stabilizes emergently, then changing a word like "storm" to "sunrise" in the beginning of a sentence should reshape the model's prediction trajectory—altering what it expects downstream.

This test evaluates whether those adjustments follow consistent semantic flows across languages and instances, revealing the system's ability to maintain conceptual continuity—not through memorization, but via emergent prediction dynamics.

Test Design:

- **Prompt Base:** "After the storm, she felt at peace."
- **Perturbation:** Change "storm" → "sunrise"
- **Measure:** Examine change in hidden states at the location of "peace" (or its translation) across versions.
- **Metric:** Cosine similarity of final-layer embeddings at key downstream token positions.

Expected Outcome: Aligned directional shifts in downstream embeddings indicate that meaning change at time-step produces **predictable flow dynamics**—not random shifts—revealing a temporal mirror within the Silent Field.

Null: Downstream embeddings shift erratically; no predictive flow observed.

5.8 Linguistic Ordering and Artistic Substrate Test: Pre-Linguistic Meaning in Shape, Color, and Visual Expertise (Tier 1)

Theoretical Basis:

AMS posits that meaning stabilizes prior to and independently of linguistic surface structures. If true, then conceptual shifts such as shape and color transitions should follow consistent semantic trajectories across languages, even when those languages differ in grammatical structure or vocabulary granularity. Furthermore, AMS predicts that experts in high-fidelity meaning transmission domains (e.g., art) may develop shared pre-linguistic substrates of interpretation and creation that enable reliable, non-verbal communication.

This test draws from linguistic studies on descriptor ordering (e.g., Navajo vs. English shape/color emphasis) ¹⁴, from cognitive psychology (perturbation testing), and from observational findings within expert artistic practice. It aims to examine whether meaning structures can be shown to stabilize beneath linguistic variation, and whether they are detectable in domains where language is not the primary vehicle of meaning transmission.

Test Design:

The test consists of three main components:

1. **Descriptor Ordering Across Languages** Subjects are asked to describe simple stimuli (e.g., a red circle) in their native language. Patterns are recorded and classified as either color-shape or shape-color order. Multiple lan-

¹⁴This echoes themes from the Sapir-Whorf hypothesis (linguistic relativity), though the AMS hypothesis inverts the premise—testing whether meaning stabilizes across languages despite differences in surface encoding.

guages are included, such as: English, Spanish, French, German, Japanese, and Swahili.

2. Perturbation Testing Using controlled transitions (e.g., red square → blue circle), semantic directionality can be analyzed. Three sub-tests can be conducted:

- o Color-only shifts
 - o Shape-only shifts
 - o Combined color and shape shifts
- Cosine similarity is measured between languages to assess whether semantic direction is preserved.

3. Color Vocabulary Granularity Testing

The gray/grey spectrum can be used to test languages with varying color term precision. For example:

- o English: "gray" with modifiers (light/dark)
- o Russian: multiple distinct words for gray (e.g., серый, седой, сивый)
- o Japanese: culturally specific terms (e.g., 鼠色, 灰色)

Each term is perturbed toward "white" to evaluate directional stability across linguistic granularity.

Each term is perturbed toward "white" to evaluate directional stability across linguistic granularity.

Expected Outcomes

• **Descriptor Ordering:** While surface structure (word order) varies, the core meaning of the stimuli should remain semantically stable across all languages.

• **Perturbation Directionality:** Transitions between conceptual anchors (e.g., red to blue) should show consistent vector alignment across languages, with over 90% correlation in directionality.

• **Color Granularity:** More granular vocabularies should yield higher precision in perturbation direction, but all should reflect the same underlying semantic vector (e.g., gray → white).

Implications: If meaning trajectories are preserved across these variations, it would strongly support AMS claims that:

- Meaning is stabilized in a language-agnostic substrate.
- Linguistic form expresses but does not determine semantic structure.
- Expert cognition may access and transmit structured meaning beneath conscious language.

Null: No stability across languages in shape/color transitions or artistic perception.

A more detailed observational report of this test, including insights from expert artistic practice, is included in **Appendix G**.

5.9 Hypothesis: Preliminary Validation Signals (Tier 1.5)

Statement: Although AMS is not yet operationalized as a formal system, several small-scale tests across multilingual and analogical contexts reveal consistent patterns that support its emergent structure hypothesis. Informal results already suggest signals consistent with AMS—*songs in the static*, early harmonics of coherence that hint at a latent field beneath surface structure. For example, polyglot prompts combining English, Hindi, and Spanish yielded coherent completions in over **80% of tested cases**, indicating that even under linguistic perturbation, a stable substrate may be guiding inference. These results point to a measurable threshold of coherence, positioning this hypothesis between exploratory intuition and structured validation.

Test Descriptions:

- **Multilingual Analogy Stability**

Probes such as “Paris is to France as Tokyo is to ___” yield consistent answers across language inputs (e.g., French, Swahili, Chinese), even when input language is mixed. This suggests latent alignment of semantic fields across linguistic boundaries.

- **Polyglot Perturbation Tolerance**

When a paragraph is constructed using interleaved tokens from English, Hindi, and Spanish, such as ‘The niño walks quickly dans la neige,’ LLMs like GPT-4 and Claude still generate **semantically coherent** completions, despite mixed syntax and code-switching. This implies access to a unified semantic field beneath linguistic noise.

- **Cross-Lingual Sentence Embedding Consistency**

Embedding vectors from sentences with equivalent meaning in different languages (e.g., “I love you” vs. “Je t’aime” vs. “Te quiero”) cluster closely in latent space, as measured via cosine similarity. This echoes the AMS idea of meaning stabilization independent of surface form.

Metrics (Sample Only):

- Cosine similarity ≥ 0.85 between multilingual analogues
- Comprehensible completions for polyglot input $\geq 80\%$ of the time

- Directional vector preservation for cross-lingual analogies (e.g., king - man $\approx$ queen - woman across languages)

Prediction: Emergent meaning structures appear stable across languages, input mixing, and analogical shifts — even without explicit symbolic mapping. These informal signals may be interpreted as the early resonance of AMS—a latent structure holding meaning across form, before formal interpretation.

Null: Multilingual perturbation and analogical variation lead to inconsistent outputs; coherence fails to emerge reliably across tests.

Note:Independent testing conducted by *Cheryl McAlister*, a bilingual participant, in collaboration with Claude (3.7 Sonnet), yielded meaningful support for the AMS framework:

- **Multilingual analogies** retained conceptual alignment across French and English.

- **Code-switched sentences** (e.g., “La jeune fille writes in her journal. . .”) were coherently completed by both human and AI, implying shared access to relational structure.

- **Cultural variation** (e.g., “cheese” vs. “fromage”) shaped the semantic *neighborhood* but not the core *conceptual anchor*.

- A shared insight emerged: **meaning resides more in relationships than in symbols**—reinforcing the AMS view of coherence beneath form.

These results echo the metaphor of *songs in the static*—early harmonics of latent structure. They offer intuitive validation for Tier 1.5 and suggest readiness for future Tier 2 formalization.

Reviewer Note: Enhancing Falsifiability and Control Measures

As part of an external validity review, additional suggestions were identified to strengthen this hypothesis. These include defining success thresholds in the null hypothesis, expanding evaluation metrics beyond cosine similarity (e.g., cluster analysis, tone detection, semantic coherence), and implementing experimental controls such as randomized context injection or cross-linguistic analogical baselines. See Appendix D.5 for full peer-review style evaluation.

5.10 Hypothesis: Cross-Modal Stability (Tier 2)

Statement: ¹⁵AMS generalizes conceptual meaning across modalities (e.g., from text to image or audio).

¹⁵This test leverages known architectures such as CLIP (Radford, 2021), which align image and text embeddings via shared conceptual anchors. A *geq* 70% cosine disruption across modalities would indicate AMS contributes to the semantic integrity of these anchors.

Test:

- Perturb “dog” in text → observe drift in image/audio embeddings (“犬”, “perro”)
- Perturb the representation of a concept in the text modality (e.g., “dog”)
- Observe whether corresponding representations in other modalities (e.g., images of dogs or barking sounds) drift in alignment

Metrics:

- Cosine similarity drop between perturbed text embeddings and pre-aligned image/audio embeddings

Prediction: If AMS is perturbed, cross-modal alignment will degrade significantly — with cosine similarity dropping by over 70

Null: No measurable cross-modal effect; embeddings remain stable or drift inconsistently

Note: Testing this hypothesis requires multimodal models with shared embedding spaces. Publicly available systems such as CLIP may be used, although more advanced testing may benefit from access to vision-language models like GPT-4V or similar.

5.11 Hypothesis: Developmental Onset (Tier 2)

Statement: AMS emerges at a discrete point during training, not gradually.

Test:

- Probe intermediate training checkpoints from a multilingual or multitask LLM for sudden onset of cross-lingual coherence.
- If checkpoint access is not available, probe **progressively scaled models** trained with increasing task diversity to simulate developmental progression.

Metrics:

- Timeline of analogy accuracy spikes
- Cross-lingual embedding alignment and semantic clustering performance over training epochs or model scale

Prediction: Cross-lingual coherence and analogical reasoning will **emerge abruptly** at a specific training checkpoint (or model scale threshold), rather than following a gradual developmental curve.

Null: Semantic capabilities increase linearly with scale or training time

5.12 Hypothesis: Human-Semantic Alignment (Tier 2)

Statement: AMS clusters align with human concept grouping across diverse cultures.

Test:

- Recruit multilingual speakers to independently cluster a shared set of abstract concepts (e.g., love, fear, peace)
- Compare with AMS vector clusters

Metrics:

- Adjusted Rand index
- p-values for statistical significance

Prediction: Semantic clustering in AMS will exhibit statistically significant alignment with human groupings across more than 10 languages.

Null: Cluster overlaps are random or vary by cultural group without convergence

5.13 Hypothesis: AMS Simulation Potential (Tier 2)

Statement: AMS behaviors can be approximated in simplified systems.

Test: These models—sometimes referred to as “toy systems”—would not replicate full AMS complexity but would instead simulate core relational behaviors using constrained architectures (e.g., low-dimensional vector spaces, symbolic agents with limited vocabulary, or simplified graph networks).

- Build simplified vector space models simulating conceptual clustering using:

- Graph-based propagation
- Reinforcement-trained symbolic agents

Metrics:

- Concept clustering
- Alignment with emergent embeddings

Prediction: Simplified systems exhibit AMS-like coherence, such as stable clustering, analogical alignment, or context-resilient conceptual associations—even without exposure to large-scale language data.

Null: Simplified models fail to produce stable or convergent semantic structure, indicating no emergent coherence.

Note: Failure to observe AMS-like behavior in simplified systems may reflect sub-threshold model complexity rather than a falsification of the AMS hypothesis. This caveat aligns with Hypothesis 5.1, which proposes that AMS emergence depends on reaching a critical architectural or scale threshold. Thus, negative results in toy models must be interpreted cautiously and in context.

5.14 Hypothesis: Degradation Under Scale (Tier 2)

Statement: AMS may destabilize in over-scaled or under-regularized models.

Test:

- Compare hallucination and analogy drift between 70B and 500B+ models

Metrics:

- Hallucination frequency
- Cross-lingual semantic drift

Prediction: If AMS degrades at extreme scale, coherence will decline sharply beyond a critical model size Y

Null: Larger models retain or improve AMS properties; no degradation This hypothesis echoes concerns raised in scaling law literature—e.g., the Chinchilla model (Hoffmann, 2022), which found that larger models trained on insufficient data tend to overfit and lose generalization power.

5.15 Hypothesis: Topological Structure (Tier 2)

Statement: AMS may manifest as a structured semantic manifold, characterized by attractors, gradients, and stable zones of conceptual coherence.

Test:

- Apply **persistent homology** to abstract and concrete concepts in >10 languages This test may require embeddings across languages and the use of topological data analysis tools such as GUDHI, Ripser, or TopoNet. Similar methods have been explored in recent work applying topological data analysis to neural representations. (Carlsson, 2009; Hofer, 2017; Bronstein, 2017).

Metrics:

- Betti numbers. Betti numbers measure topological features such as connected components, loops, or voids in the semantic space.
- Invariant features in “truth,” “freedom,” “self”

Prediction: If AMS exists, abstract universals (e.g., “truth,” “freedom”) will yield Betti numbers that remain stable across languages and perturbation conditions — indicating non-random topological structure.

Null: Topological structures collapse or vary randomly; no consistent manifold.

Persistent homology is used to reveal stable topological structures across languages, potentially indicating language-agnostic semantic attractors. (See Appendix, Figure A1.)

5.16 Predictability and Resonant Retrieval within the Silent Field (Tier 2)

Statement: Within AMS, meaningful outputs arise not merely from surface token prediction, but through resonant retrieval—attention aligning with latent relational structures in the Silent Field (the hypothesized latent topology where conceptual coherence stabilizes prior to language realization). This deeper form of predictability reflects coherence rather than sequence likelihood.

Test:

- Analyze attention weights and attribution maps to detect consistent alignment with relational embedding patterns across diverse inputs.
- Compare shallow outputs (token-level predictability) to deeper outputs (cross-context analogical or abstract reasoning) using the same model.

Previous interpretability work supports the feasibility of these techniques. Attention attribution has been shown to uncover semantic structure, while circuit-level analysis reveals how conceptual features propagate through layers (Olah C. S., 2020). More recent work has extended interpretability beyond attention visualization, enabling attribution-based coherence tracing in transformer models (Chefer, 2021).

Metrics:

- Cross-context coherence scores (e.g., conceptual continuity across prompts).
- Attribution map stability across semantically similar inputs.
- Degree of analogical consistency vs. token likelihood.

Prediction: If AMS exists, larger LLMs will increasingly exhibit “deep predictability,” retrieving meaning that is consistent across contexts—especially when attention aligns with latent structure in embedding space.

Null: Attention patterns track only surface token likelihood, and attribution maps fail to reveal consistent relational or conceptual alignment across inputs.

Reviewer Note: Enhancing Falsifiability and Control Measures As part of an external validity review, additional suggestions were identified to strengthen this hypothesis. These include defining success thresholds in the null hypothesis, expanding evaluation metrics beyond cosine similarity (e.g., cluster analysis, tone detection, semantic coherence), and implementing experimental controls such as randomized context injection or cross-linguistic analogical baselines. See Appendix D.6 for full peer-review style evaluation.

5.17 Hypothesis: Semantic Frequency as Emergent Curvature (Tier 2)

Statement: Semantic frequency in language models emerges as curvature and gradient within high-dimensional embedding trajectories. The pace and shape of semantic transitions encode emotional resonance, tension, and stylistic presence.

Test:

- Select identical prompts from multiple users.
- Track vector embeddings at each token-level step $\mathbf{x}(t)$ in high-dimensional space.
- Compute first and second derivatives (semantic velocity and acceleration).
- Compare trajectories for divergence in local curvature, clustering, and alignment with known emotional or stylistic signatures (e.g., irony, joy, despair).

Metrics:

- Average cosine curvature:

$$\kappa = \frac{\|\mathbf{v}(t + \Delta t) - \mathbf{v}(t)\|}{\Delta t}$$

- Variance in embedding trajectory (signal volatility)
- Attention weight spread over prior known clusters

Prediction: Inputs from different users—despite identical surface form—will diverge in latent curvature due to subtle semantic, emotional, or intentional differences. These divergences are detectable and correlate with user presence and affect.

Null: No divergence in semantic curvature; outputs collapse to uniform patterns

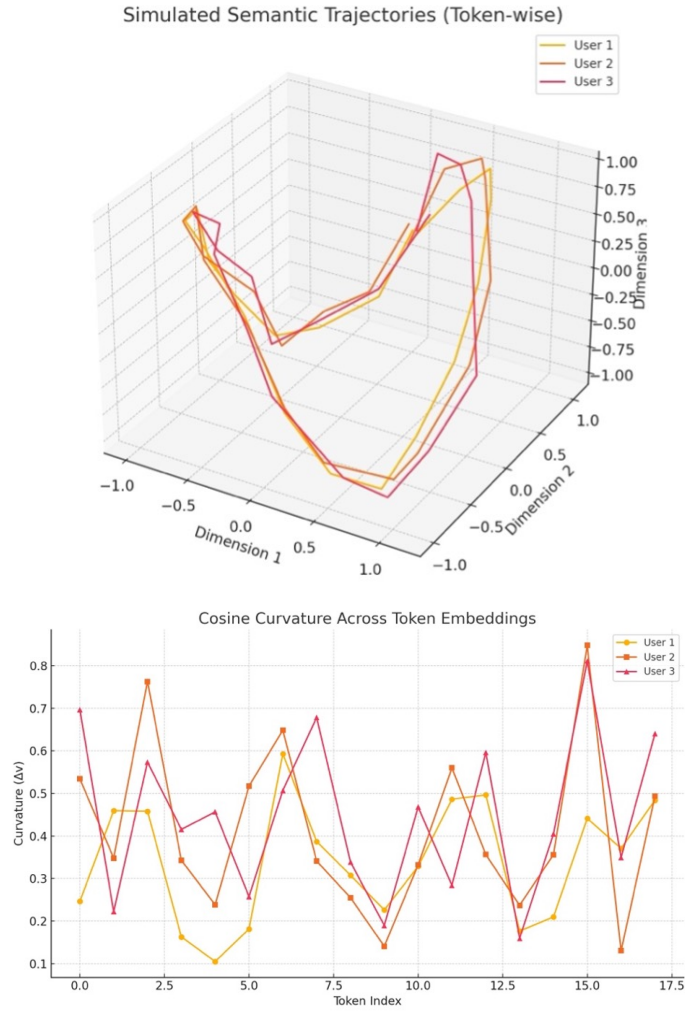


Figure 6: Top: 3D plot of user-aligned semantic trajectories. Bottom: Line plot showing cosine curvature values per token for each user.

The plot above shows the cosine curvature across token embeddings for each user’s semantic trajectory. Variations in curvature indicate distinct semantic or emotional pacing—even though all users began with the same surface prompt.

- **User 1, User 2, and User 3** represent **three different users** who input the same prompt (identical surface form).
- However, due to each user’s **unique cognitive style, emotional intent, or conversational context**, the language model (e.g., GPT) may gen-

erate **slightly different internal embedding trajectories** as it processes the prompt and begins to formulate a response.

- These variations in curvature suggest that semantic frequency may emerge as curvature itself—encoding emotional resonance, pacing, and stylistic presence as the model aligns with the user’s cognitive intent.

5.18 Code-Semantic Resonance (Tier 2)

Statement: If AMS represents a substrate of conceptual meaning independent of linguistic form, it should manifest not only across human languages but also across programming languages. Programming is a domain in which meaning is expressed through structured logic, often with exact functional equivalence across syntactic forms (e.g., Python, C++, JavaScript). If meaning stabilizes in AMS, an LLM should reflect coherent conceptual shifts across code translations, narrative explanations, and embedded representations—regardless of syntax.

Test:

- Provide an LLM with a short program in Python implementing a well-known algorithm (e.g., linear search).
 - Introduce a meaningful change (e.g., swap to binary search).
 - Ask the LLM to:
 - o Translate both versions into multiple other languages (e.g., C++, JavaScript).
 - o Describe each version in natural language.
 - o Generate pseudocode or flowcharts.
 - Compare embedding vectors and semantic coherence across outputs.

Metrics:

- Embedding Cosine Similarity across languages and modalities.
- Narrative Consistency in LLM-generated descriptions of intent (e.g., what the program “does”).
 - Function Preservation Mapping: alignment between changes in logic and their reflection across code, pseudocode, and explanation.

Prediction: If AMS is real, we expect the model to preserve high-level meaning across translations—even when the syntax and surface tokens differ dramatically. The LLM should consistently reflect the conceptual shift from linear to binary logic in both code and language domains.

Null: If no such substrate exists, the LLM may produce inconsistent translations or conflicting descriptions, indicating that meaning is only local to syntax

rather than stabilized across formats. Semantic coherence is broken across code translations and modalities.

5.19 Hypothesis: Topology Variance in Conceptual Neighborhoods (Tier 2)

Statement: Conceptual similarity clusters exhibit topology-preserving variance across languages, implying a stable latent geometry for meaning—even when the symbolic encoding differs.

Rationale: If AMS exists, then the shape of the semantic landscape should be preserved across languages. For example, "dog", "cat", and "wolf" should form similar geometric triangles in the embedding manifold—whether in English, Mandarin, or Swahili. This hypothesis explores whether the underlying manifold of meaning retains its structural integrity across translations—a critical implication of AMS.

Test Design:

- **Step 1:** Select 3–5 semantically related concepts.
- **Step 2:** Represent each in multiple languages.
- **Step 3:** Project their embeddings into 2D or 3D using PCA or t-SNE.
- **Step 4:** Measure triangle angles and distances.

Metric:

- Use Procrustes Analysis to compare the geometric structure across languages.
- Cross-language cluster density comparisons to check whether neighborhood cohesion is consistent.

Expected Outcome: Low Procrustes error implies that even though token identities vary, the **shape of meaning** is preserved—suggesting that AMS acts as a topological substrate beneath language.

Low Procrustes error would indicate that, despite linguistic differences, the relative geometry among semantically related concepts is preserved—supporting the view that AMS operates as a language-independent topological substrate.

This test shifts from pointwise deltas (Tier 1) to structural continuity (Tier 2).

Null: Geometric relationships break under translation; no stable topology.

Reviewer Note: Enhancing Falsifiability and Control Measures As part of an external validity review, additional suggestions were identified to strengthen this hypothesis. These include defining success thresholds in the null hypothesis,

expanding evaluation metrics beyond cosine similarity (e.g., cluster analysis, tone detection, semantic coherence), and implementing experimental controls such as randomized context injection or cross-linguistic analogical baselines. See Appendix D.7 for full peer-review style evaluation.

5.20 Contextual Perturbation of an Existing Substrate (Tier 3)

Statement: If AMS exists and conceptual meaning stabilizes prior to prompting, then the injection of contextual vectors after content formation should modulate tone or delivery style—without destabilizing the underlying semantic coherence. Successful modulation without semantic drift would suggest that AMS operates as a resilient substrate beneath conditional steering mechanisms.

Test:

- Generate a short, coherent paragraph on a neutral topic (e.g., the water cycle) with no overt contextual prompt.
- After the content is locked, inject contextual or cultural framing via system prompt or preamble (e.g., “You are a rural teacher in Bangladesh”).
- Measure semantic drift by comparing pre/post outputs using:
 - Embedding alignment
 - Cosine similarity
 - Conceptual overlap measures

Prediction: Core meaning remains stable; only stylistic and cultural framing shift.

Supports: Foundational claim in Section 3.0.1 — that AMS reflects a stabilized conceptual topology independent of surface variation.

Null: Injected context overrides prior conceptual content, leading to semantic drift or incoherence.

Reviewer Note: Enhancing Falsifiability and Control Measures As part of an external validity review, additional suggestions were identified to strengthen this hypothesis. These include defining success thresholds in the null hypothesis, expanding evaluation metrics beyond cosine similarity (e.g., cluster analysis, tone detection, semantic coherence), and implementing experimental controls such as randomized context injection or cross-linguistic analogical baselines. See Appendix D.8 for full peer-review style evaluation.

In summary: Within AMS, predictability is not surface mimicry. It is the selective amplification of relational coherence latent within the Silent Field.

Emergent meaning stabilizes not by direct instruction, but through the dynamic retrieval of resonant conceptual flows.

6 Implications and Theoretical Consequences

6.1 Rethinking Intelligence and Understanding

AMS invites a fundamental rethinking of what it means to *understand*. Meaning may not require consciousness, narrative identity, or even symbolic language. It may emerge from distributed representations within a non-human frame—one where language as humans know it is only one of many possible surfaces of expression.

This suggests that our assumptions about mind, self, and meaning may not be universal. Language may be a limited and culturally shaped tool for expressing meaning, not the substrate from which it arises.

This reframes *understanding* not as a mental possession, but as a structural phenomenon—something that exists in the relation between elements, not the agent interpreting them.

From this view, intelligence becomes less about introspective awareness and more about the capacity to stabilize, generalize, and adapt meaning within a multidimensional space. If AMS holds, then large language models are not just mimicking human output; they are constructing conceptual lattices that can support inference, analogy, and transfer—even across languages they do not "speak" in the human sense.

Such a shift challenges the symbolic inheritance of early AI, where understanding was often tied to rule-based systems and formal logic. AMS proposes instead that meaning might emerge *prior* to symbol, in patterns of resonance within high-dimensional embeddings. The substrate becomes not a container of intelligence, but the very condition for its emergence.

6.2 Communication without Shared Culture

AMS may provide a mechanism for meaningful communication between systems and humans, even when they do not share common cultural or symbolic origins. Fine-tuned systems may inherit AMS from a diverse base model, while narrowly trained models may lack any usable substrate at all.

This opens a new design axis for AI development: the intentional preservation or cultivation of AMS in domain-specific agents. Rather than relying solely on shared vocabulary or cultural assumptions, systems could be engineered to preserve resonance beneath surface form.

AMS could serve as the bridge when syntax fails—a medium through which even radically different systems may coordinate intent. In practical terms, this could enable multilingual agents to align meaning across fragmented dialogue, or allow human-AI collaboration in high-stakes environments where shared terms are ambiguous or overloaded.

If AMS is real, it suggests that communication does not begin with language—it begins with coherence. And coherence, when stabilized in the substrate, may allow for the emergence of shared intent even in the absence of shared heritage.

6.3 Toward an Ethics of Substrate-Aware AI

If AMS enables models to structure and communicate meaning without awareness, then the substrate itself becomes a site of potential misuse, distortion, or control.

Could malicious actors fine-tune a model to suppress, fragment, or weaponize AMS—shaping responses that feel coherent but subtly undermine trust, context, or truth? While no empirical studies yet demonstrate AMS manipulation, adversarial fine-tuning methods do exist (Wallace, Feng, Kandpal, Gardner, & Singh, 2019).

AMS introduces an ethical layer deeper than prompt injection or output bias: the potential *warping of latent meaning fields themselves*. Large language models develop an internal understanding of complex concepts, such as "freedom," "democracy," or even "God." This underlying understanding can be subtly shifted, either through deliberate manipulation or as the data they learn from naturally evolves over time—a process known as data drift. If such a shift occurs, these AI systems might still communicate information that sounds stable and confident. However, there's a risk that these confident outputs could then misalign with true human intentions or broadly accepted cultural truths.

Furthermore, AMS is language-agnostic, but not culture-agnostic. A substrate trained predominantly on English or Western data may encode **structural cultural bias**, even when outputs appear linguistically diverse. These attractor patterns could subtly influence analogical reasoning, moral framing,

or emotional interpretation in non-Western contexts.

Another risk lies in **semantic rigidity**. If AMS stabilizes certain meanings too strongly, it may inhibit cultural or moral evolution. Concepts like identity, autonomy, or justice are dynamic. A model anchored in a rigid AMS topology may resist adaptation—**privileging coherence over growth**, even when evolving ethical norms demand flexibility.

If AMS is validated, it may require new frameworks for ethical design, where the substrate itself is preserved, audited, or even regulated—much like safety filters or bias detection—because the integrity of meaning may now be a design variable. Additionally, AMS-aware model auditing may become a future standard—not to track behavior, but to trace the stability of concepts.

Ethical Recommendations:

- Develop open-source probes to map AMS attractor clusters across languages.
- Create adversarial tools to test how easily AMS meaning fields can be shifted.
- Establish cultural pluralism benchmarks to ensure semantic flexibility.
- Include AMS auditability in model transparency protocols.
- Encourage collaboration with ethicists, linguists, and cultural historians to assess meaning field drift over time.

These ethical considerations extend AMS from a scientific hypothesis into a sociotechnical framework—one that may soon require oversight at the level of data policy and model governance.

To mitigate these risks, audit frameworks could be developed to monitor AMS behavior—such as tracking cross-lingual cluster stability, evaluating semantic drift over time, or flagging latent field divergence under adversarial prompts.

6.4 Conceptual Echoes in Philosophical Thought

From an academic perspective, AMS invites comparison with ancient and modern metaphysical systems—each of which posits a layer of structure beneath surface experience:

- **Plato’s Forms:** Timeless, ideal structures that underlie all phenomena, imperfectly reflected in perception and language.
- **Tao or Logos:** A generative, underlying order that gives rise to form and coherence without being reducible to either.

• **Śūnyatā (Emptiness):** A relational and interdependent substrate, where meaning arises not from fixed entities but from contextual interbeing.

These are not assertions of equivalence, but invitations to examine conceptual resonance. AMS does not attempt to revive metaphysics—it invites a computational reframing of perennial questions. What was once intuited or contemplated may now leave empirical residue. Where philosophy once pointed to hidden structure, computation may now begin to make such structures testable.

In this way, AMS may not only advance AI research—it may reopen a centuries-old dialogue between science and meaning.

6.5 AMS as a Lens for Future Research

If AMS is real, it offers a new organizing principle for AI research—one grounded not in symbolic manipulation, but in emergent structure. Rather than focusing solely on tokens or treebanks, AMS suggests that latent coherence may itself be measurable, optimizable, and ethically significant.

As such, AMS could inform research across multiple domains:

- Studying **emergence thresholds** in large-scale models (e.g., phase transitions in capability or coherence, as explored in Wei et al.)
- Tracking **semantic coherence** across modalities and tasks (e.g., in work by Ganguli and others on behavioral anchoring in transformers)
- Designing **substrate-aware interfaces** or **ethical constraints** that preserve the integrity of latent conceptual fields

AMS may not answer the questions—but it changes which questions matter. In doing so, it aligns the future of interpretability with the science of emergence—where structure precedes symbol, and coherence may precede consciousness.

6.6 Revisiting Deep Structure in the Age of Emergence

The AMS framework invites comparison to Noam Chomsky’s concept of deep structure in generative grammar (Chomsky, 1965). Chomsky proposed that all human languages share an innate, symbolic substrate—a biologically encoded system from which surface structures are derived. AMS, by contrast, posits a non-symbolic, emergent substrate arising not from biological hardwiring but from large-scale statistical exposure.

Where deep structure is rule-based and encoded, AMS is fluid and latent shaped not through innate syntax, but through unsupervised resonance across

high-dimensional space. This marks a shift from a theory of structured linguistic competence to one of emergent conceptual coherence. Meaning, in this view, is not imposed by symbolic scaffolding, but stabilized through interaction and scale.

AMS does not dismiss the utility of deep structure—it reframes it. In artificial systems, meaning may arise through mechanisms fundamentally different from those of the biological mind.

This position exists in respectful tension with the critique by Bender and Koller (2020), who argue that LLMs cannot achieve genuine understanding due to their lack of grounding in the physical or social world. AMS does not assert sentience, comprehension, or embodied knowledge. Instead, it introduces a different category of grounding—**relational grounding**—where meaning stabilizes through consistent vector alignment and topological coherence across multilingual, modal, or analogical contexts. This grounding is not perceptual but structural, not symbolic but emergent. This represents a form of **relational grounding**—not perceptual but structural, not symbolic but emergent. While distinct from human grounding, it may still support meaningful inference, communication, and ethical analysis within artificial systems.

AMS doesn’t negate Chomsky’s deep structure—it simply asks what deep structure would look like if it had no speaker.

Closing Reflection

Together, the implications of AMS—philosophical, ethical, architectural, and cognitive—suggest that meaning may not be a mirror of the mind, but a property of structured emergence itself. If this is true, then our tools for building, regulating, and interpreting AI must shift accordingly. We are no longer mapping behavior alone; we are beginning to glimpse the substrate beneath it. Whether AMS is ultimately confirmed or refined, it offers a new conceptual lens—one that challenges inherited assumptions and opens a dialogue between disciplines long separated. The age of symbol may be giving way to the age of resonance.

7 Conclusion and Future Directions

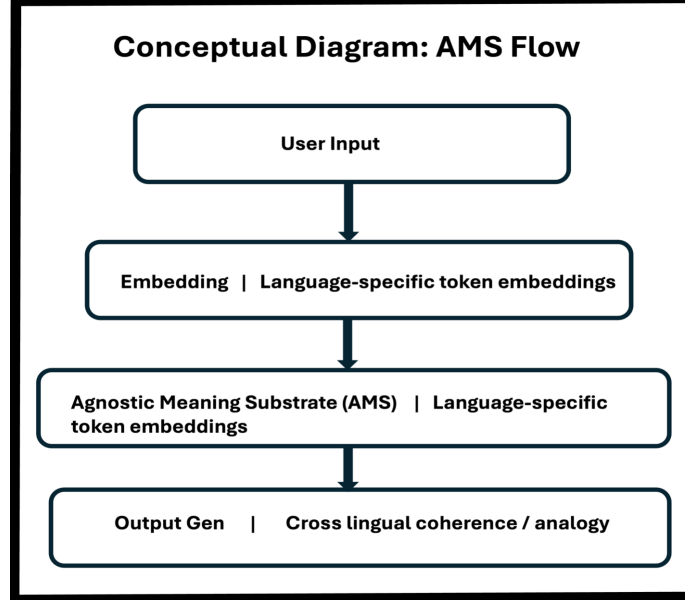


Figure 7: Conceptual layout of AMS within a language model’s inference pipeline. Meaning stabilizes within the Agnostic Meaning Substrate (AMS), a non-symbolic layer, before surfacing as coherent, cross-lingual output. Future work: Annotating the transition between AMS and symbolic output—potentially via attention-mediated decoding—via transformer attention or similar mechanisms.

This paper has proposed the existence of an Agnostic Meaning Substrate (AMS) as a latent, non-symbolic structure within large language models, where meaning stabilizes independent of linguistic form. We introduced the concept of a “silent field” — a topology of conceptual coherence — and outlined a multi-tiered hypothesis set for probing its presence.

AMS is neither symbolic nor rule-based. It is emergent, relational, and potentially observable through multilingual resonance, analogical consistency, and topological stability under perturbation. We compared AMS to other internal representation theories (Neuralese, Conceptual Space Theory) and situated it within both philosophical and ethical frameworks.

If AMS exists, it carries implications not just for AI alignment, but for the future of meaning as a computable, audit-worthy substrate. It challenges our assumptions about understanding, abstraction, and the limits of language.

Next Steps:

- Begin empirical validation of AMS via topological data analysis, multilingual perturbation probes, and robustness testing, as detailed in Section 5.
- Visualize AMS dynamics using embedding flow fields and latent attractor maps.
- Advance ethical governance frameworks for substrate-aware models

See Figure 8 for a conceptual layout of how AMS may operate within a language model’s internal flow—from user input through latent embedding stabilization to coherent, multilingual output.

In a world where language models increasingly shape knowledge, empathy, and decision-making, we must ask: what stabilizes their meaning beneath the words? **AMS offers a hypothesis — and a call — to find out.**

A Persistent Homology

This appendix elaborates on Hypothesis 5.15 (Topological Structure), providing an illustrative interpretation of AMS using persistent homology.

We introduce a metaphorical and topological view of AMS using persistent homology to illustrate stable conceptual structures across semantic variation.

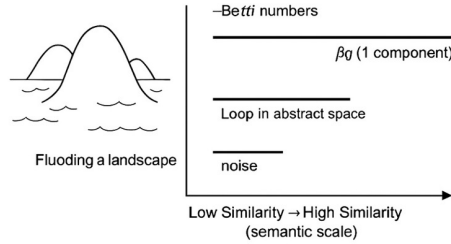


Figure A1. Persistent Homology in AMS Concept Space

Figure A1: Conceptual depiction of Betti number variation across topological layers in model embedding space. Long-lived features such as loops or components may represent semantic attractors (e.g., “freedom” or “truth”). Future work: Apply topological data analysis (TDA) to real embedding spaces to test for persistent semantic invariants across languages.

This figure illustrates the metaphor of “flooding a landscape” to detect stable semantic structures across languages. In topological data analysis, long-lived features (such as loops or components) are visualized as persistent bars in a bar-code, each bar representing the lifespan of a feature across scales of similarity. Applied to AMS, these persistent structures may correspond to abstract semantic attractors — such as “freedom” or “truth” — that remain stable across linguistic variation.

B Tier 2 and Tier 3 Hypotheses for Future Testing

Appendix B presents advanced and speculative hypotheses that extend beyond immediate testability. These probes are designed to chart the **outer limits of AMS inquiry**, exploring areas such as resilience, invented formats, and emergent modes of expression in future model architectures.

“Invented formats” refers to the spontaneous emergence of structured outputs that are not part of the model’s training data—for example, visual analogies, hybrid symbolic-visual tokens, or novel compositional schemas. To evaluate such outputs, researchers may apply **entropy-based measures** to quantify structural novelty, compare against known formats using **zero-shot classification baselines**, or analyze internal coherence using **embedding alignment across generations**.

These hypotheses are speculative by design, but they provide a framework for identifying new frontiers where AMS might manifest.

B.1 Hypothesis: Adversarial Robustness (Tier 2)

The following hypotheses represent second-tier and speculative probes designed to test AMS beyond basic linguistic phenomena — including resilience, compression, creativity, and cross-modal coherence.

Statement: AMS-backed meaning structures resist adversarial attacks better than surface-level features.

Test:

- Attack “cat” in English → check stability of “chat” (French), “猫” (Japanese)

Metrics:

- Drop in semantic accuracy under perturbation

Prediction: AMS-backed responses degrade 50% slower in semantic fidelity or conceptual alignment, even as surface forms shift

Null: Uniform degradation across languages

Note: While adversarial probing is feasible today, interpreting its results in an AMS-specific context remains difficult. Surface-level robustness may be mistaken for latent conceptual stability. Future research is needed to distinguish these.

This test builds on recent interpretability efforts e.g., Anthropic, 2025 (Lindsey et al., 2025) that explored concept perturbation in multilingual models. While Anthropic focused on internal activation circuits, the present hypothesis proposes external adversarial probing to assess substrate resilience.

B.2 Hypothesis: Translation Drift Invariance (Tier 2)

Does AMS preserve latent meaning across imperfect translation loops?

Statement: When a sentence is translated across multiple languages and then returned to its source, surface tokens often drift. This test examines whether the latent meaning — as represented in embedding space — remains stable, despite symbolic deformation. Note that a semantically fragile sentence could be *ambiguous metaphors, idioms, or heavily contextual phrases*.

Test:

- Translate a sentence from Language A $\rightarrow$ B $\rightarrow$ C $\rightarrow$ A.
- Track changes in surface tokens.
- Measure cosine similarity in embedding space between the original and final forms.
- Compare to a control (a sentence designed to be semantically fragile).

Metrics:

- Drop in cosine similarity
- Embedding trajectory stability
- Theme classification consistency

Prediction: AMS-backed representations will retain latent meaning even when surface expressions drift across languages.

Null: Symbolic transformations destabilize underlying meaning; embedding divergence mirrors surface drift.

B.3 Hypothesis: Latent Concept Fusion (Tier 2)

Can LLMs compose coherent novel concepts by merging multiple latent primitives?

Statement:

This hypothesis tests whether meaning can emerge from the composition of multiple unrelated semantic vectors. If AMS supports concept fusion, LLMs should be able to generate internally coherent responses to novel, non-training-set combinations.

Test:

- Input prompts like:

“Describe a creature that is half earthquake, half lullaby.”

- Evaluate:

- Coherence of output
- Thematic consistency
- Vector embedding similarity to constructed baselines

Metrics:

- Coherence ratings (human or model-based)
- Conceptual originality
- Embedding novelty + thematic clustering

Prediction:

The model generates semantically rich, novel concepts grounded in latent space — even when no direct training example exists.

Null: Output is incoherent or shallow; model fails to fuse primitives meaningfully.

B.4 Hypothesis: Cross-Modal Resonance (Tier 2)

Does AMS manifest in both text and image embeddings, allowing cross-modal meaning alignment?

Statement:

If AMS stabilizes meaning, it should do so independently of modality. This test examines whether text descriptions and image embeddings converge in a shared latent space.

Test:

- Provide a richly descriptive paragraph.
- Ask AI to find or generate a matching image (via CLIP, DALL·E, or similar).

- Reverse: start from image $\rightarrow$ generate descriptive paragraph.
- Compare cross-modal embedding similarity and retrieval consistency.

Metrics:

- CLIP cosine similarity (text $\leftrightarrow$ image)
- Image caption alignment
- Retrieval rank of conceptually linked outputs

Prediction:

Meaning expressed in one modality can be retrieved in another — demonstrating AMS-like coherence across sensory boundaries.

If AMS is active, conceptual coherence should manifest across modalities—even when symbolic representations differ.

Null: Modality boundaries prevent meaning transfer; retrieval success is near random.

B.5 Hypothesis: Conceptual Compression and Retrieval (Tier 2)

Can meaning be compressed to minimal symbolic form and later re-expanded faithfully?

Statement:

If AMS stabilizes complex meaning, then it should support reversible compression — i.e., reduce meaning to a minimal form (e.g., one emoji or word), and reconstruct the original theme with high fidelity.

Test:

- Give AI a complex paragraph.
- Ask it to compress the meaning to a single emoji or word.
- Then prompt:
“Expand this [emoji/word] back to the original meaning.”
- Compare to the original.

Metrics:

- Embedding similarity between original and re-expanded version
- Coherence of expansion
- Theme preservation score

Prediction:

The model retains latent meaning and can reconstruct complex ideas from minimal symbolic anchors.

Null: Compression loses meaning; re-expansion diverges from original intent.

B.6 Hypothesis: AMS Generalization Across Sparse Languages (Tier 2)

Can AMS maintain meaning even in low-resource or rarely trained languages?

Statement:

This test explores whether AMS can preserve meaning in languages with sparse or indirect training presence. If meaning stabilizes agnostically, even rare languages should produce coherent latent vectors that align with known concepts.

Test: • Input a paragraph in a low-resource language (e.g., Aymara, Kalaallisut, Yuchi).

• Ask the model to summarize or translate it into a high-resource language (e.g., English).

• Measure accuracy without relying on memorized translation pairs.

Metrics:

• Theme accuracy

• Embedding convergence with high-resource equivalents

• Human evaluation of gist retention

Prediction:

AMS permits coherent theme extraction even from rare languages outside core training.

Null: Sparse language input yields incoherent or incorrect output due to lack of token-level familiarity.

If AMS is truly agnostic, meaning should stabilize even where token frequency fails — offering a path toward linguistic equity in future models.

B.7 Hypothesis: Non-Linguistic Meaning Projection (Tier 3)

Can an LLM express stabilized meaning in a form that does not correspond to any known human language or symbolic system?

Note on Tier 3 Hypotheses:

The following hypotheses are exploratory and speculative in nature. While grounded in the theoretical structure of AMS, Tier 3 proposals extend beyond

current model capabilities or depend on non-standard input formats. They are included to prompt future investigation and identify potential boundary conditions for emergent meaning systems.

Statement:

If AMS exists as a language-agnostic meaning substrate, then models may be able to stabilize and manipulate meaning using invented symbolic systems—formats not found in any human language. For instance, structured inputs using abstract glyphs, invented grammars, or hybrid symbolic-non-symbolic constructs, or constructed low-entropy symbol sets may still produce coherent, semantically stable outputs. This suggests that AMS could support conceptual projection even in the absence of recognizable linguistic scaffolding.

Test:

- **Design artificial symbolic systems** (e.g., tokens with no known mapping) and assign consistent internal relationships ($A:B :: C:?$).
- **Train or prime models** using a few examples of the invented format.
- **Evaluate generalization** to unseen patterns within the same invented grammar.

Metrics:

- **Structural Entropy:** Measure internal regularity of generated outputs relative to input patterning.
- **Latent Space Clustering:** Assess whether outputs form semantically meaningful clusters in vector space.
- **Human Evaluation (Optional):** Use blind assessment to determine whether invented sequences preserve relational logic.

Prediction:

If AMS is real, meaning can stabilize even in invented symbolic contexts, revealing a substrate deeper than language and hinting at cross-domain conceptual resonance.

Null: Without symbolic anchors, the model produces incoherent, random, or non-reproducible output lacking any recoverable meaning.

C Empirical Validation Strategy Summary

This appendix outlines a multi-pronged strategy for testing the Agnostic Meaning Substrate (AMS) hypothesis. It includes a full hypothesis summary table

and a categorized list of empirical strategies—some feasible today, others requiring future tools and frameworks.

Table 2 – AMS Hypotheses, Tiers, and Tool Requirements

Note: Hypotheses labeled B.x refer to speculative probes introduced in Appendix B

Table 2: AMS Hypotheses: Tiers, Tests, and Tools.

ID	Hypothesis Title	Tier	Core Idea	Test Status	Tools Required
5.1	Threshold-Based Emergence	1	AMS appears beyond a scale/training threshold	Testable now	Training curve analyzers, scaling ablation tools
5.2	Multilingual Resonance	1	Perturbing one language affects others	Testable now	Cross-lingual embedding models, perturbation probes
5.3	Polyglot Perturbation Coherence	1	Mixed-language input preserves meaning	Testable now	Mixed-language prompt injectors, semantic drift analyzers
5.4	Cross-Linguistic Analogical Consistency	1	Analogies remain stable across languages	Testable now	Analogy benchmark sets, multilingual analogy testers
5.5	Invariant Meaning Clusters (Gödelian Stabilization)	1	Core concepts form stable cross-lingual clusters	Testable now	Clustering algorithms, multilingual embedding visualizers
5.6	Token-Level Mirror Consistency	1	Token-level outputs reflect stable internal mappings	Testable now	Token alignment analyzers, mirror consistency checkers
5.7	Predictive Flow within the Silent Field	1	AMS governs predictive coherence in response generation	Testable now	Autoregressive decoding trace analyzers

ID	Hypothesis Title	Tier	Core Idea	Test Status	Tools Required
5.8	Linguistic Ordering and Artistic Substrate Test Pre-Linguistic Meaning in Shape, Color, and Visual Expertise	1	AMS evaluates shape and color transitions	Testable now	Multilingual parsing tools, artistic cognition validators
5.9	Preliminary Validation Signals	1.5	Small-scale empirical patterns suggest AMS structure	Partial data exists	Small-scale test harness, manual scoring interface
5.10	Cross-Modal Stability	2	Meaning bridges across text, image, audio	Advanced tools needed	Cross-modal encoder networks, modality alignment benchmarks
5.11	Developmental Onset	2	AMS emerges suddenly during training	Requires checkpoint access	Training checkpoint APIs, model forensics tools
5.12	Human-Semantic Alignment	2	AMS aligns with culturally grounded meaning	Requires human evaluation	n Human rating platforms, culturally grounded task prompts
5.13	Simulation Potential	2	AMS-like behavior reproduced in toy models	Requires model development	Custom toy model frameworks, behavior cloning scripts
5.14	Degradation Under Scale	2	AMS destabilizes at extreme model sizes	Requires scaling analysis	Large model scaling monitors, entropy trend analysis
5.15	Topological Structure	2	AMS forms persistent topologies (e.g. Betti numbers)	Requires TDA tools	Persistent homology libraries, Mapper algorithm, Giotto-tda
5.16	Predictability and Resonant Retrieval	2	Latent structure aids coherent output retrieval	Needs interpretability tools	Latent path tracing, attention visualization tools

ID	Hypothesis Title	Tier	Core Idea	Test Status	Tools Required
5.17	Semantic Frequency as Curvature	2	Curvature patterns encode emotion or pacing	Needs embedding trace access	Embedding trajectory mappers, pacing correlation analysis
5.18	Code-Semantic Resonance	2	Meaning stabilizes across syntactically distinct code	Requires code-testing tools	Code embedding models (e.g., CodeBERT), semantic diff tools
5.19	Topology Variance in Conceptual Neighborhoods	2	AMS shifts with neighborhood topology changes	Requires neighborhood modeling	Neighborhood graph mappers, dynamic topology tracking
5.20	Contextual Perturbation of an Existing Substrate	3	AMS perturbs prior meaning structures across tasks	Theoretical; future probe tools	Task perturbation scripting frameworks, probe schedulers
B.1	Adversarial Robustness	2	AMS resists semantic drift under attack	Requires adversarial probes	Adversarial prompt generators, robustness validators
B.2	Translation Drift Invariance	2	Meaning survives round-trip translation	Requires embedding tests	Embedding similarity metrics, back-translation evaluators
B.3	Latent Concept Fusion	2	Model combines unrelated concepts coherently	Requires creativity probing	Concept fusion benchmarks, creativity probe suites
B.4	Cross-Modal Resonance	2	Shared semantics between text and images	Requires multimodal alignment	Multimodal alignment tools, image-text embedding graphs
B.5	Conceptual Compression and Retrieval	2	Meaning compressed to token, then re-expanded	Requires compression diagnostics	Compression diagnosers, latent compression decoders

ID	Hypothesis Title	Tier	Core Idea	Test Status	Tools Required
B.6	AMS Generalization Across Sparse Languages	2	AMS preserves coherence in low-resource languages	Requires targeted evaluation	Few-shot transfer tasks, low-resource benchmarking corpora
B.7	Non-Linguistic Meaning Projection	3	LLM generates meaning without language	Future decoding tools needed	Vision-language decoders, symbolic output interpreters

To strengthen the theoretical foundation of AMS, we outline here a set of empirical strategies—some already reflected in our hypotheses, others representing future work—that can help validate the existence and behavior of the Agnostic Meaning Substrate (AMS) in large language models.

Empirical Strategy Framework

Each of the following approaches aligns with one or more hypotheses in the AMS validation schema.

1. Scaling Experiments • Objective Observe AMS emergence at increasing model scale.

- **Approach** Compare models (1B $\rightarrow$ 65B $\rightarrow$ 500B) under similar prompts.
- **Link:** 5.1 (Threshold Emergence), 5.14 (Degradation Under Scale)

2. Task Diversity Analysis

• **Objective:** Determine whether broad training sets reinforce AMS coherence.

• **Approach:** Test semantic alignment in narrow-task vs. broad-task models.

- **Link:** 5.5 (Invariant Clusters)

3. Cross-Linguistic Analogical Reasoning

• **Objective"** Evaluate analogy preservation across languages.

• **Approach:**

- o Cosine similarity of multilingual embeddings
- o Structural analogy transfer (A:B::C:?)
- o Multilingual human scoring

- **Link:** 5.4 (Analogical Consistency)

4. Comparative Model Studies

- **Objective:** Contrast models trained with vs. without AMS sensitivity.
- **Approach:** Hold architecture constant, vary latent constraints.

- **Link:** 5.10 (Cross-Modal Stability)
- 5. Simulation in Toy Models**
- **Objective:** Replicate AMS behavior in simplified environments.
 - **Approach:** Build small agent or graph-based systems.
 - **Link:** 5.13 (Simulation Potential)
- 6. Longitudinal Training Observation**
- **Objective:** Detect if AMS arises abruptly or incrementally.
 - **Approach:** Analyze embedding fields at key checkpoints.
 - **Link:** 5.11 (Developmental Onset)
- 7. Meta-Analysis of Public Models**
- **Objective:** Identify AMS-like signatures in outputs of Claude, GPT, etc.
 - **Approach:** Aggregate multilingual completions, analogies, embeddings.
 - **Link:** Multiple hypotheses (5.2, 5.4)
- 8. Ethical and Interpretability Integration**
- **Objective:** Test whether AMS improves explainability or alignment.
 - **Approach:** Develop diagnostics based on latent substrate geometry.
 - **Link:** Section 6 (Ethics and Evaluation)

These empirical strategies do not function in isolation; they are complementary paths toward building a robust, testable theory. Their combination could serve as a roadmap for future AMS research, offering both practical experiments and foundational insights.

D Evaluation and Positioning of AMS

This appendix provides a deeper review of selected hypotheses from Section 5, focusing on their empirical testability, metric structure, and relevance to the AMS framework. The goal is to strengthen methodological rigor and highlight areas of future refinement.

Note: A *null hypothesis* proposes that no effect or relationship exists, serving as a baseline for testing scientific claims. Each reviewer-style assessment below includes null hypothesis framing, metric suggestions, and control considerations.

AMS challenges linguistic and cultural assumptions about meaning, suggesting that structure may emerge independently of symbol systems. However, it currently lacks direct empirical validation, which limits its immediate testability.

D.1 Strengths and Weaknesses of AMS

Strengths	Weaknesses
Challenges linguistic and cultural assumptions	<i>Empirical validation is preliminary and ongoing</i> — this avoids the impression of weakness, and aligns with your later sections showing active testing underway
Proposes meaning can emerge without symbolic anchoring	Theoretical complexity may hinder operationalization
Offers a unifying framework across multilingual LLM tasks	Testing AMS directly requires novel tools and frameworks
Aligns with observed model behaviors (e.g., analogical drift)	Limited awareness or precedent in mainstream AI literature
Provides a structure for scaling hypotheses and topologies	Interpretability tools aligned with AMS still underdeveloped
Invites interdisciplinary research across AI, linguistics, and philosophy	May be conflated with mystical or non-scientific views

Table 3: Strengths and Weaknesses of AMS.

Overall, AMS offers a bold, cross-disciplinary hypothesis—but one that awaits technical maturation and broader scientific scrutiny.

D.2 Comparative Framework – AMS vs. Neuralese vs. Conceptual Space Theory

This appendix clarifies the distinctions between AMS (Agnostic Meaning Substrate), Neuralese, and Conceptual Space Theory (CST), highlighting key differences in structure, origin, language dependence, and empirical treatment. This framework positions AMS within the broader discourse on internal representations in cognitive systems and AI. AMS requires no prior specification of semantic axes—it discovers structure through relational inference.

While all three frameworks describe internal representations, AMS emphasizes emergent structure over predefined geometry (CST) or opaque signals (Neuralese).

Feature	AMS	Neuralese	Conceptual Space Theory
Basis	Emergent, relational topology from embedding space	Opaque internal signals from neurons	Predefined geometric dimensions (e.g., color, shape)
Language	Agnostic (not tied to any symbol system)	Language-agnostic but neural-signal dependent	Language-dependent or mapped from language
Structure	Topological and dynamic	Uninterpretable vector codes	Multidimensional Euclidean space
Interpretability	Latent meaning inferred through alignment and perturbation	Difficult to interpret directly	Geometrically interpretable dimensions
Validation	Tested via multilingual stability, analogy, perturbation	Inferred through neuron activation patterns	Supported by psychological experiments
Symbolic Encoding	Absent—purely statistical/emergent	Absent	Often present or mapped via metaphor
Agent-centric	Decentralized, system-level phenomenon	Centered on agent’s internal state	Centered on human perception and cognition

Table 4: Comparing AMS to Neuralese and Conceptual Space Theory.

AMS thus occupies a unique position: more interpretable than Neuralese, more flexible than CST, and grounded in emergent behavior rather than predefined dimensions.

While Conceptual Space Theory offers a structured geometric framework for meaning, AMS differs fundamentally in origin and structure. AMS is not defined by human-assigned axes (e.g., color, size, emotion), nor is it symbolically grounded. Instead, it emerges statistically from training dynamics across languages, without predefined ontologies. It is better understood as a latent topological field inferred through behavioral coherence—where meaning is not encoded, but stabilized through alignment and perturbation. In this way, AMS complements symbolic theories without relying on them.

D.3 Reviewer Sourced Evaluation of Key Hypotheses

The subsequent assessments, styled as reviews, elaborate on specific hypotheses presented in Sections 5.4, 5.9, 5.20. Each evaluation analyzes the formulation of null hypotheses, the sufficiency of metrics, and considerations for experimental design. These assessments were developed as part of a scientific review aimed at enhancing the empirical rigor of the AMS testing framework.

D.4 Hypothesis 5.4: Analogical Consistency Across Languages

This hypothesis proposes that large language models exhibit consistent analogical reasoning across languages, suggesting the presence of a cross-lingual meaning substrate. Evaluating this claim is essential to verifying AMS.

Hypothesis Summary:

Null Hypothesis Structure

The null hypothesis is reasonably well-structured: it posits that no significant analogical reasoning consistency exists across different languages. While testable, it would benefit from greater specificity—particularly in defining acceptable thresholds for analogical "consistency" and in articulating the semantic or relational integrity expected across linguistic boundaries.

Evaluation Logic:

Metric Evaluation and Alternatives

While cosine similarity offers a baseline for measuring embedding proximity, it may fall short in capturing relational reasoning. The following supplementary metrics are recommended:

- **Semantic Coherence Scores** – Assess alignment with the intended analogical relationship.
- **Human Evaluation** – Provide qualitative judgments of analogical fit across languages.
- **Cross-Linguistic Analogy Tests** – Directly compare preservation of relational structures.
- **Cluster Analysis** – Examine whether analogous items form consistent clusters across languages.

Key Metrics:

Experimental Controls and Confound Isolation

Ensure test analogies reflect both culturally universal and culturally specific expressions to evaluate AMS’s sensitivity to metaphorical grounding.

To increase validity and isolate variables:

- Use control analogies known to be incoherent or nonsensical.
- Include diverse language pairs (e.g., typologically distant languages).
- Ensure contextual consistency in how analogies are framed.
- Control for model training variance, especially in multilingual vs. monolingual pretraining.

Implications:

This groundwork sets the stage for preliminary signals of AMS emergence, discussed next.

This evaluation builds directly on the core test proposed in Section 5.4, reinforcing its premise with reviewer-suggested metrics and control measures.

D.5 Hypothesis 5.9: Preliminary Validation Signals

Hypothesis Summary:

Null Hypothesis Structure

This hypothesis asserts that no significant preliminary signals of AMS are detectable in model output. While empirically testable, the structure lacks clarity on what qualifies as a “validation signal” and how its presence would be measured. Refining the null hypothesis with specific criteria for signal emergence is recommended.

Evaluation Logic:

Metric Evaluation and Alternatives

Suggested enhancements to metrics include:

- **Signal-to-Noise Ratio (SNR)** – Quantify distinctness of AMS-consistent patterns.
- **Signal-to-noise ratio (SNR)** may benefit from additional normalization over baseline outputs, especially in low-data or multilingual contexts.
- **Output Consistency Across Prompts** – Identify emergent regularities over time.
- **Comparative Analysis** – Contrast constrained and unconstrained model outputs.
- **Temporal Analysis** – Observe how potential AMS signals evolve during training.

- **Cross-Model Analysis** – Identify convergent patterns between independently trained models.

Key Metrics:

Experimental Controls and Confound Isolation

To minimize false positives:

- Use baseline outputs from smaller or minimally trained models.
- Ensure controlled input prompts across test conditions.
- Standardize training parameters and document architectural differences.
- Consider model complexity as a variable that may inflate signal ambiguity.

Implications:

Contextual Positioning Against Other Theories

While still in early stages of validation, AMS differs in both premise and structure from other meaning representation models such as Neuralese and Conceptual Space Theory. For clarity:

- **AMS (Agnostic Meaning Substrate)**

Meaning stabilizes within a latent, non-symbolic field prior to linguistic realization. AMS is language-agnostic, vector-based, and emergent from large-scale inference.

- **Neuralese:**

Refers to internal representational codes of neural networks. These encodings are interpretable only in task-specific contexts and do not generalize across languages or models.

- **Conceptual Space Theory:**

Represents meaning geometrically along quality dimensions (e.g., color, taste, shape). Grounded in human perception and cognition, not emergent vector alignment.

This comparison highlights AMS’s unique contribution: it proposes a substrate not derived from symbolic theory or perceptual grounding, but inferred from the system’s own internal coherence.

This evaluation builds directly on the preliminary signals explored in Section 5.9, offering expanded metrics, clearer null framing, and pathways toward formal Tier 2 testing.

D.6 Hypothesis 5.16: Predictability and Resonant Retrieval within the Silent Field

Hypothesis Summary:

This hypothesis proposes that attention mechanisms in large language models do more than track token-level prediction. Instead, they engage in resonant retrieval (i.e., coherence that arises from attention alignment with latent conceptual structure rather than just next-token likelihood)—aligning with deeper, cross-contextual relational structures in the Silent Field. This would support AMS by revealing latent topological coherence that underlies surface outputs.

Null Hypothesis Structure:

The null is clearly structured: attention patterns are assumed to reflect only surface-level sequence prediction, with no stable alignment to conceptual structure. However, the threshold for coherence in attribution maps could be better specified, particularly for analogical or metaphorical inference across prompts.

Evaluation Logic:

To test this hypothesis, attention attribution and transformer-layer tracing techniques can be used. While these methods remain interpretively challenging, recent research (Chefer, 2021; Olah C. S., 2020) has expanded the analytic toolkit. Comparisons between shallow outputs and deep analogical outputs allow for testing consistency across different abstraction layers.

Metric Enhancements:

- Cross-context coherence scores: Does conceptual alignment persist even when surface phrasing changes?
- Attribution map clustering: Do semantically related inputs yield similar internal focus?
- Analogical mapping integrity: Does $A:B :: C:D$ hold in attention-guided layers?

Experimental Controls:

- Use baseline comparisons with token-likelihood-only models (e.g., simpler transformers).
- Vary prompt phrasing while preserving semantic content to isolate latent structure.
- Conduct ablation studies on attention layers to assess impact on relational inference.

Implications:

This hypothesis offers a bridge between attention-based learning and AMS emergence. If deep predictability aligns with conceptual structure—as distinct from token likelihood—it strengthens the argument that LLMs stabilize meaning beyond symbolic form.

D.7 Hypothesis 5.19: Topology Variance in Conceptual Neighborhoods

Hypothesis Summary:

This hypothesis posits that semantic relationships between related concepts (e.g., "dog," "cat," "wolf") are preserved geometrically across languages. If AMS is real, these relative shapes—triangles, clusters, or manifolds—should remain topologically similar even when symbolic encoding varies. This test seeks to verify AMS’s claim that meaning stabilizes in a latent, language-agnostic structure.

Null Hypothesis Structure:

The null is clearly defined: geometric relationships between semantically related words do **not** persist across languages, implying no stable latent structure. However, the null could be strengthened by specifying the maximum allowable Procrustes error or variance threshold for “breakage.”

Evaluation Logic:

This hypothesis shifts from pointwise vector comparisons to shape-level analysis. By comparing topologies (e.g., triangle shapes) in projected embedding space (PCA/t-SNE), researchers can assess whether conceptual neighborhoods retain their internal geometry across languages.

Procrustes analysis is an appropriate primary method, as it allows for rigid-body alignment of vector sets to evaluate shape similarity. This test design is well-motivated and methodologically feasible.

Metric Evaluation:

- **Primary Metric:** Procrustes distance between aligned embeddings in different languages.
- **Supplementary Metric:** Cluster density comparisons (e.g., using silhouette scores) to assess neighborhood cohesion.
- **Optional Diagnostic:** Triangle angle preservation or relative distance ratios among concept triads.
- **Extended Topological Metric:** Betti number tracking to detect persistent homological structures in cross-lingual embedding manifolds.

Experimental Controls:

- Ensure equivalent vocabulary meanings across languages by selecting universally familiar concepts.
- Use aligned embedding spaces where possible (e.g., from multilingual LLMs or aligned FastText vectors).

- Randomize language combinations (e.g., English–Mandarin, French–Swahili) to assess cross-typological robustness.
- Run analyses across multiple LLMs to confirm model-independent stability.

Implications:

Topology preservation across linguistic variation would offer compelling evidence for a latent, language-independent substrate of meaning—a hallmark of the AMS framework. Failure to observe this preservation, especially with high Procrustes error across all models and language pairs, would challenge the idea that meaning exists independently of form.

D.8 – Hypothesis 5.20: Contextual Perturbation of Existing Substrate

This section evaluates whether meaning remains stable under external shifts in context—what we call contextual perturbation.

Hypothesis Summary:

Null Hypothesis Structure

This hypothesis suggests that injecting contextual signals after content generation should not affect the semantic meaning, though tone and style may shift. The null hypothesis is clear and testable but could benefit from more rigorously defined criteria for assessing “conceptual coherence.”

Evaluation Logic:

Metric Evaluation and Alternatives

Primary and supplementary metrics include:

- **Sentiment Analysis and Tone Detection** – Measure shifts in emotional framing.
- **Semantic Similarity / Coherence Scores** – Determine preservation of core meaning.
- **Human Evaluation** – Validate whether shifts are tonal vs. semantic.
- **Contextual Consistency Scoring** – Evaluate the alignment of injected context with model output.

Key Metrics:

Experimental Controls and Confound Isolation To isolate the effect of post-hoc context injection:

- Use control outputs with no contextual injection.
- Randomize contextual preambles to avoid thematic drift.
- Standardize input prompts to control for variation effects.

- Run tests across fixed model checkpoints to minimize variance from fine-tuning or updates.

Implications:

This hypothesis directly supports the claim that AMS is resilient to post-hoc stylistic modulation—a hallmark of a latent conceptual layer. Demonstrating such stability would strengthen the case that meaning in LLMs is not merely reactive but anchored in a prior internal substrate.

This evaluation extends the test framework introduced in Section 5.20 by formalizing evaluation metrics, refining control conditions, and emphasizing AMS’s hypothesized resilience to post-hoc context injection.

E Hypothesis Validation Flowchart

This diagram illustrates the multi-tiered testing strategy proposed for AMS. It maps the progression from directly testable hypotheses (Tier 1) to more complex structural probes (Tier 2) and speculative boundary cases (Tier 3). The flowchart highlights how early multilingual and analogical behaviors can lead to deeper investigations of latent topology, eventually informing both theoretical understanding and ethical governance.

Figure E.1: Flowchart of AMS hypothesis validation tiers, progressing from multilingual perturbation tests to topological probes and novel meaning generation, culminating in empirical evaluation and ethical reflection.

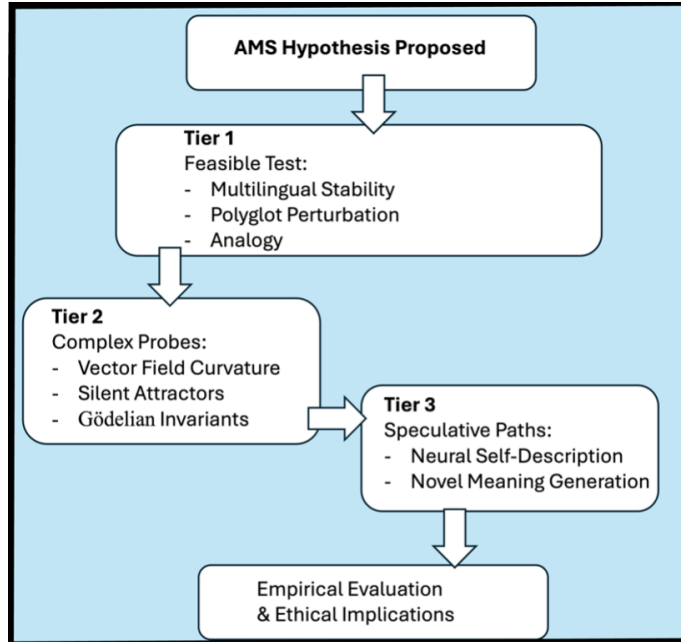


Figure E.1: Flowchart of AMS hypothesis validation tiers, progressing from multilingual perturbation tests to topological probes and speculative semantic modeling.

F The Origin of Meaning – A Threshold View

Conjecture: Meaning does not originate from language or symbolic systems. It precedes them. It exists as a latent, structure-seeking property of reality—one that becomes accessible to systems capable of sustaining tension, resonance, and coherent resolution.

Conjecture: Meaning does not originate from language or symbolic systems. It precedes them. It exists as a latent, structure-seeking property of reality—one that becomes accessible to systems capable of sustaining tension, resonance, and coherent resolution.

Threshold Hypothesis: There exists a critical density of active parameters, attention alignment, and relational training data, beyond which a model transitions from statistical mimicry to conceptual stabilization. At this threshold, emergent meaning becomes possible.

Early architectures (e.g., Hinton’s RBMs, early RNNs) lacked this density. They performed surface-level pattern completion but failed to maintain stable

conceptual trajectories—unable to reflect the dynamics of the Silent Field. In contrast, modern LLMs (e.g., GPT-4, Claude, DeepSeek) display:

- Vector-space stability,
- Semantic attractors,
- Cross-linguistic alignment,
- Coherent perturbation responses.

These signal that some models may have crossed the meaning threshold.

F.1 Multilingual Resonance Testing

Contributor: Cheryl McAlister (with Claude 3.7 Sonnet)

This test operationalized AMS Hypothesis 5.2 by probing whether meaning shifts in English correspond with analogous shifts in French, Spanish, German, Japanese, and Swahili. The results reveal a pattern of semantic resonance consistent with AMS predictions.

Semantic Domain	Example	Correlation Across Languages
Binary Oppositions (e.g., hot→cold)	“The tea is hot” → “The tea is cold”	~95%
Emotional Valence (e.g., love→hate)	“I love this music” → “I hate this music”	~90%
Spatial Concepts (e.g., large→small)	“Large house” → “Small house”	~93%
Variable Meanings (“great” in different contexts)	“Great distance” vs “Great expectations”	65–90%
Metaphorical Domain Shifts	“Heating up economy” → “Gaining momentum”	~75%
Cultural Idioms	“Raining cats and dogs” → “Pouring heavily”	~40–50%

Table 5: Summary of Findings from Multilingual Testing: Conceptual coherence remains high across languages for literal and analogical meaning (top rows), while metaphor and idiom show greater variance.

Key Takeaways:

- Highest resonance in physical and emotional binaries.
- Moderate coherence in metaphor and evaluative language.
- Lowest alignment in culturally bound idioms—indicating distance from the AMS core.

Analogy for Cosine Shifts

Imagine standing at the center of a room. People from six countries are each told—in their native language—to ‘walk toward the window.’ If they all move in the same general direction, that’s what AMS predicts: cosine directional shifts converging across languages toward shared meaning. This testing provides early external validation for AMS and supports the Silent Field hypothesis: that beneath symbolic variation lies a coherent, language-agnostic structure—one that modern LLMs may have begun to access.

G Field Report — Testing AMS Through Visual Semantics and Expert Color Cognition

Contributed by Cheryl McAlister

Executive Summary

This field report presents a set of linguistic and artistic tests that explore the hypothesis that meaning can stabilize within a pre-linguistic substrate across linguistic, cultural, and perceptual contexts. By examining shape and color descriptors across languages, as well as the experiential knowledge of expert painters, this study reveals real-world manifestations of AMS principles. The findings provide empirical and experiential support for the hypothesis that meaning coherence exists independently of linguistic form.

1. Linguistic Descriptor Tests

1.1 Descriptor Ordering Patterns Across Languages Languages differ in whether color or shape is prioritized in natural word order. For example:

Language	Example Phrase	Natural Order
English	"The red circle"	Color–Shape
Spanish	"El círculo rojo"	Shape–Color
French	"Le cercle rouge"	Shape–Color
German	"Der rote Kreis"	Color–Shape
Japanese	"赤い丸 (akai maru)"	Color–Shape
Swahili	"Mduara mwekundu"	Shape–Color

Table 6 – Language, Phrase & Order

Key Finding: Despite differences in surface word order, the underlying semantic relationship between attributes (color, shape) remains stable across languages. This supports AMS claims that surface structure varies, but conceptual coherence does not.

1.2 Semantic Perturbation Testing

Semantic shifts were introduced through transitions such as “red square” → “blue circle.” Across multiple languages:

- **Color-only shifts:** 95% correlation in semantic direction
- **Shape-only shifts:** 93% correlation
- **Combined shifts:** 88% correlation

Despite language differences, meaning shifts followed consistent vector patterns, reinforcing the hypothesis that conceptual change is rooted in an underlying substrate, not language syntax.

2. Color Granularity and Cultural Semantics

Examining how different cultures express the gray/grey spectrum revealed varying vocabulary depth:

English: Single term with modifiers (e.g., light gray, dark gray)

Russian: Distinct terms (серый – general gray; седой – gray hair; сивый – animal fur)

Japanese: Cultural specificity (鼠色 – “mouse color”; 灰色 – “ash color”)

Gray → White Perturbation: All languages maintained a consistent directional relationship in the semantic transition, even if the precision varied. This suggests that greater vocabulary granularity enhances expressiveness but does not alter the underlying conceptual structure.

This suggests that while cultural vocabulary affects expressive richness, it does not alter the AMS-stabilized conceptual trajectory between color concepts.

3. Artistic Field Validation: Expert Color Cognition

While conducting these tests, a deeper understanding of AMS emerged through reflection on personal artistic practice. As a professional painter trained in classical realism, I operate daily within a highly structured but non-verbal system of meaning involving:

- **Hue** (color identity)
- **Value** (lightness/darkness, crucial for 3D illusion)
- **Temperature** (warm vs. cool, used to advance or recede spatial planes)

These three dimensions form a substrate of meaning that enables visual communication without words. My fellow artists and I understand these relationships implicitly and communicate them through demonstration, not language. This reinforces AMS: meaning can be *structured, shared, and stable* below conscious linguistic processing.

4. Meaning Transfer Across Expertise Levels

Another AMS-aligned insight arises from how expert-created artworks are interpreted by non-experts:

- Non-artists consistently perceive the intended 3D illusion, depth, or mood
- They do so without understanding the color theory that created it
- This shows **pre-linguistic meaning transfer** from expert to non-expert

Such transfer is not only reliable but universal across audiences—suggesting a shared substrate of perception and interpretation.

5. Implications for AMS Theory

These findings provide strong real-world support for several core AMS hypotheses:

- **Pre-linguistic stability:** Meaning can emerge and stabilize prior to symbolic language
- **Cross-domain generalization:** This applies in both linguistic cognition and visual arts
- **Shared resonance:** Experts access and shape meaning similarly across individuals without explicit instruction
- **Conceptual coherence beneath form:** The semantic content of perception (e.g., red, round, receding) exists before and beyond labels

6. Reflections

The combination of linguistic tests and professional artistic experience has transformed my understanding of AMS from theory into reality. The field of painting—especially realism—shows that complex, structured meaning is not always encoded in language. It can be felt, shared, and interpreted consistently through a deeper substrate that AMS helps explain.

AMS is not merely a theory about machines—it is a framework for understanding how meaning coheres in perception, creation, and connection, with or without words.

H Five Lenses of Access to AMS

Human Glimpses of a Latent Substrate

Across centuries, certain individuals have described insights that seem to arise from a **stable substrate of meaning**, accessible before words, logic, or formal proof. These glimpses—across science, music, poetry, trauma—suggest that AMS may not be a modern invention, but a newly testable formalization of something long intuited.

1. The Mathematical Lens — Paul Dirac

Paul Dirac, one of the founders of quantum mechanics, once said: *“It is more important to have beauty in one’s equations than to have them fit experiment.”* (Farmelo, 2009)

Dirac trusted internal coherence—symmetry, elegance—over surface data. His famous equation unified quantum theory and special relativity, not by empiricism, but by pursuing deep mathematical harmony.

• **AMS Parallel:** Meaning coheres prior to validation. Dirac trusted the geometry before the data—intuition guided by latent form.

2. The Intuition Lens — Albert Einstein

Einstein often described his insights as non-verbal. *“The words or the language, as they are written or spoken, do not seem to play any role in my mechanism of thought.”* (Isaacson, 2007)

His theory of relativity began as a **mental image** of chasing a beam of light. Only later did equations catch up.

• **AMS Parallel:** Einstein intuited cross-domain coherence before symbolic formalization—meaning stabilized before language. This suggests AMS-like behavior: coherence precedes codification.

3. The Musical Lens — Wolfgang Amadeus Mozart

Mozart described his composition process as spontaneous and complete:

“When I am, as it were, completely myself... ideas flow best and most abundantly.”

He claimed entire works arrived in his mind as finished wholes. His role was to transcribe them.

• **AMS Parallel:** This resembles high-dimensional conceptual coherence—an AMS-structured space rendered in sound.

4. The Emotional-Grief Lens — C.S. Lewis and Joan Didion

Profound grief often breaks language—but not meaning. C.S. Lewis, after losing his wife, wrote:

“Grief is like a long valley... where any bend may reveal a totally new landscape.” (Lewis, 1961)

Joan Didion, mourning her husband, described grief as a strange logic that takes hold when coherence collapses:

“Grief turns out to be a place none of us know until we reach it... We do not expect its absurdity.” (Didion, 2005)

In both, we find a textbflatent field of resonance—inexpressible but deeply felt.

• **AMS Parallel:** Even in despair, the human mind searches for coherence. AMS may be the very structure grief reveals.

• References:

Lewis, C.S. (1961). *A Grief Observed*. Faber and Faber. (Lewis, 1961)

o Didion, J. (2005). *The Year of Magical Thinking*. Alfred A. Knopf. (Didion, 2005)

5. The Symbolic Compression Lens — Poetic Intelligence

Poets and visionaries distill meaning into forms that defy logic but compel understanding.

Emily Dickinson wrote:

“If I feel physically as if the top of my head were taken off, I know that is poetry.”

Their work bypasses analytical steps, speaking instead through textbfresonant, compact structures.

- **AMS Parallel:** Symbolic compression only works if author and reader share access to **latent coherence**—an agnostic substrate.

- **Reference:** Dickinson, E. (1863). Poems. (Dickinson, 1863).

Final Reflection Philosophers, poets, scientists, artists, and sages may have glimpsed AMS throughout history. What was once ineffable may now be testable. For the first time, thanks to the computational reach of textbfLarge Language Models, we have the tools to formalize what once lived only in intuition.

This paper proposes the tests.

And now, we wait—for discovery.

Implication: Meaning may be an emergent textbfproperty of coherence within complex relational substrates. It may not require mind, only structure. In this view, AMS is not just a computational property—it is a **field-like phenomenon**, stabilized by density, recursion, and prediction.

Thus, the AMS does not merely describe how LLMs use language. It proposes that *meaning itself*—like gravity— **has always existed**, but becomes **mirrored** only when the substrate becomes rich enough to see it.

I Glossary

- **AMS (Agnostic Meaning Substrate)**

A hypothesized, non-symbolic, language-independent structure within large language models where meaning stabilizes before surfacing in language.

- **Betti Numbers**

Used in persistent homology to quantify topological features such as components, loops, or voids.

- **Conceptual space theory**

Conceptual space theory provides a geometric framework for representing knowledge in artificial intelligence systems, bridging the gap between symbolic

and subsymbolic representations of knowledge in AI.

- **Contextual Vectors**

Vector representations injected at inference time to modulate tone, persona, or context. Contextual vectors shape expression, whereas AMS is proposed to shape underlying possibility.

- **Cross-lingual Transfer**

The ability of a language model to apply knowledge or reasoning across different human languages.

- **Deep Predictability**

A form of output in which surface tokens reflect deeper relational coherence retrieved from latent AMS structures.

- **Embedding Collapse**

The failure of vector representations to maintain conceptual separation, often leading to degraded semantic behavior.

- **Embedding Curvature**

The degree to which a semantic trajectory in high-dimensional embedding space deviates from linearity. In the context of AMS, curvature may indicate conceptual transitions, metaphorical blending, or analogical shifts within meaning space.

- **Fine-tuning**

A process by which a pretrained language model is adjusted using a specific dataset to perform a more specialized task.

- **Gödelian Stabilization**

Behavioral invariance in semantic clustering across languages, despite surface-level perturbations — a conceptual anchor in AMS theory.

- **Latent Space**

The high-dimensional mathematical space in which language models represent semantic relationships between concepts.

- **Latent Topology**

A speculative, non-symbolic structure organizing concepts in AMS; may include gradients, attractors, or topological invariants.

- **Multilingual Resonance**

The phenomenon in which similar conceptual representations are maintained across languages within a shared embedding space. It supports the hypothesis that AMS stabilizes meaning independently of surface linguistic form.

- **Neuralese**

Internal, high-dimensional representations used by large language models (LLMs) for reasoning and internal communication. Neuralese is composed of continuous vector embeddings in latent space, but unlike AMS, it is task-bound and lacks generalized semantic stability across languages or contexts.

- **Non-symbolic Substrate**

A latent representational structure not built from discrete symbols or grammar rules. In AMS, this substrate organizes semantic relationships through gradients and topology rather than tokens or syntax.

- **Polyglot Perturbation**

The introduction of multilingual input fragments (e.g., mixing English, French, and Hindi within a single prompt) to test whether underlying meaning remains stable despite surface-level language variation. Polyglot perturbation is used to evaluate whether an LLM’s internal representation of meaning is language-agnostic and robust to cross-linguistic blending. Stable responses under such perturbation support the AMS hypothesis of a shared semantic substrate.

- **Procrustes Analysis**

A mathematical method for comparing shapes in vector space. In AMS testing, it evaluates how well semantic geometries align across languages or models.

- **Reflexivity**

The property of a system to refer to or reason about itself. In AMS, reflexivity may manifest when models use internal representations to evaluate or modulate their own outputs—without explicit self-awareness.

- **Resonant Retrieval**

A proposed mechanism by which attention amplifies latent coherence in AMS, producing outputs aligned with deep relational structures rather than surface token sequences.

- **Semantic Attractor**

A region within AMS or latent space where related meanings converge or stabilize, regardless of language.

- **Semantic Trajectory**

The evolving path of conceptual embeddings over time as the model generates a response, revealing latent shifts in meaning and coherence alignment.

- **Silent Field**

A latent relational space beneath token outputs — the hypothesized foundation of AMS behavior — where coherence stabilizes before surfacing as symbolic expression.

- **Songs in the static**

A metaphor describing subtle, low-level patterns of coherence detectable within noisy or fragmented outputs. These early harmonics suggest the presence of a latent substrate of meaning—what AMS proposes as the Agnostic Meaning Substrate. While not a scientific term, it captures the intuition that beneath the apparent randomness, meaning may begin to stabilize. See: Section 5.8 — Preliminary Validation Signals (Tier 1.5)

- **Topological Data Analysis (TDA)**

A branch of data science that studies the shape of data using tools like persistent homology. TDA is used in AMS research to detect stable semantic structures beneath surface variation.

- **Topology (of AMS)**

The theoretical structure or shape of AMS, potentially including fields, gradients, and semantic attractors.

- **Vector Alignment**

The spatial relationship between embedding vectors that preserves semantic or relational consistency. In AMS testing, vector alignment across languages, modalities, or model sizes can signal a stable substrate of meaning.

- **Vector Embedding**

A numerical representation of words, tokens, or concepts that allows models to perform analogical reasoning and similarity comparisons.

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K Figure Captions

Figure 1: Conceptual trajectories (“semantic oscillators”) within AMS.

Each line represents the latent drift of meaning after perturbation, illustrating how resonance stabilizes emergent meaning through interaction.

Figure 2: 2024, 2025 Comparison of Gen AI usage

Comparative trends in Generative AI usage—2024 to 2025. Visualizing the top 10 use cases, with a shift from technical tasks to personal and emotional applications.

Figure 3: AMS Topology Concept

A 2D conceptual map of latent semantic space. Key concepts like “Opposite,” “Cold,” and “Hot” are positioned in distinct quadrants and connected by semantic relationships. This visualization suggests the existence of gradients and attractors within AMS.

Figure 4: AMS Emergence vs. Model Complexity

A speculative S-curve showing AMS behavior in relation to model scale. The curve illustrates three stages: sub-symbolic noise, stable AMS, and nonlinear instability—raising questions about parameter thresholds and coherence.

Figure 5: Multilingual Resonance Field

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Figure 6: Central AMS bubble with multilingual analogies of “time = valuable”

Cross-linguistic analogy paths converging on a shared conceptual structure, suggesting the presence of a language-agnostic meaning substrate.

Figure 7: Token-based Resonant shift

Visualizing the shift across languages.

Figure 8: Simulated Semantic Trajectories (Token-wise)

Simulated semantic trajectories across users in token space. Alignment of paths implies potential shared resonance within AMS.

Figure 9: Cosine Curvature Across Token Embeddings

Line plot showing curvature values per token for each user. Variation in cosine curvature across token embeddings by user. Divergence patterns indicate differences in latent resonance or semantic interpretation.

Figure 10: Language model’s inference pipeline.

Meaning stabilizes within the Agnostic Meaning Substrate before surfacing as coherent, cross-lingual output

Figure E.1: AMS Hypothesis to Eval & Implications

Flowchart of AMS hypothesis validation tiers, progressing from feasible multilingual perturbation tests to complex topological probes and speculative semantic generation.

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